Machine Learning-Based Detection of Telomerase Reverse Transcriptase Promoter Mutations from Routine H&E Histology in Bladder Cancer, Validated by Droplet Digital PCR Assays



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DEPARTMENT OF BIOCHEMISTRY AND MOLECULAR BIOLOGY
FACULTY OF BIOLOGICAL SCIENCES
UNIVERSITY OF DHAKA
DHAKA-1000, BANGLADESH

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Abbreviations

AI – Artificial Intelligence

ANOVA – Analysis of Variance

AUC – Area Under the Curve

BCG – Bacillus-Calmette–Guérin

BLCA – Bladder Cancer

BMI – Body Mass Index

bp – Base Pair

CNN – Convolutional Neural Network

CLS – Classification Token

ddPCR – Droplet-Digital Polymerase Chain Reaction

DNA – Deoxyribonucleic Acid

DPX – Distrene-Plasticiser-Xylene

ETS – E26-Transformation-Specific (Transcription Factor Family)

FFPE – Formalin-Fixed Paraffin-Embedded

FGFR3 – Fibroblast Growth Factor Receptor 3

GABP – GA-Binding Protein

GC – Gemcitabine-Cisplatin

H&E – Hematoxylin and Eosin

HEX – Hexachloro-Fluorescein

LOO-CV – Leave-One-Out Cross-Validation

MIBC – Muscle-Invasive Bladder Cancer

MIL – Multiple Instance Learning

NCI – National Cancer Institute

NF-κB – Nuclear Factor Kappa-B

NGS – Next-Generation Sequencing

NMIBC – Non-Muscle-Invasive Bladder Cancer

OD – Optical Density

PCR – Polymerase Chain Reaction

TERTp – TERT Promoter

RNA – Ribonucleic Acid

ROC – Receiver Operating Characteristic

SD – Standard Deviation

SVM – Support Vector Machine

TERT – Telomerase Reverse Transcriptase

TNM – Tumor-Node-Metastasis

TURBT – Transurethral Resection of Bladder Tumor

UTI – Urinary Tract Infection

VAF – Variant Allele Frequency

ViT – Vision Transformer

WSI – Whole-Slide Image

Abstract

Telomerase Reverse Transcriptase promoter *(TERTp)* mutations are the most frequently occurring somatic genetic alterations in bladder cancer. The detection of *TERTp* mutations is quite challenging due to the requirement for expensive molecular sequencing technologies, which may not always be available in resource-limited settings, such as in Bangladesh. This study pursued two complementary aims: first, to determine *TERT*p mutation frequency distribution in our cohort using highly sensitive droplet digital PCR (ddPCR); and second, to develop a deep-learning-based pre-screening tool that uses the ddPCR mutational data to detect *TERT*p mutation from routine H&E-stained histological images. The cohort used for this study comprised 77 patients with histologically confirmed bladder cancer. The tumor biopsies were cut using a microtome for histological diagnosis using the H&E staining and a section of the freshly cut tissues were used for tumor DNA extraction. ddPCR was performed on the tumor DNA for *TERTp* mutation detection. The ddPCR data analysis detected *TERT*p mutation in 39% of the tumor samples, with C228T as the predominant subtype (73.3%). In the second part of the study, a pre-screening machine learning model was built for the detection of *TERTp* mutations from the routine histology slides obtained from the H&E images. For this purpose, a transformer-based pathology foundation model was developed. A total of 120 features were selected from the biopsy images using ANOVA F-statistics and incorporated into a Gradient Boosting classifier, with histological grade retained as a covariate. The classifier was evaluated by Leave-One-Out cross-validation. The final model achieved an AUC of 0.7674, a sensitivity of 0.9, and a specificity of 0.43 for *TERTp* mutation detection at the standard classification threshold. The model is able to correctly detect 20 wild-type patients in the dataset, which can result in a 26% reduction in molecular testing workload. In subtype-specific analysis, C250T mutations were detected more accurately (AUC 0.8419) than C228T (AUC 0.7369), potentially reflecting the additional NF-κB signalling involved in C250T-mediated *TERT* activation. In grade-stratified analysis, the model achieved an AUC of 0.7944 in high-grade tumours but near-random performance (AUC 0.5256) in low-grade cases. This study demonstrates the use of conventional histological slides to detect *TERT*p mutation without requiring expensive molecular sequencing techniques, providing a cost-effective triage tool in resource-limited settings.

Keywords: Machine Learning, TERT promoter mutation, ddPCR assay, bladder cancer.

Section 1
Introduction

1.1. Background

Cancer has emerged as the second leading cause of death worldwide, surpassed only by cardiovascular disease. Globally, around 20 million new cancer cases were diagnosed, and 9.7 million deaths resulted from cancer. Lung cancer has been the most frequently occurring cancer type, comprising 12.4% of all cancers and causing 18.7% of all cancer-related deaths worldwide. According to this estimate, around one in every 5 people develop cancer in their lifetime, and one in 11 of them dies due to this. It is estimated that by 2050, cancer cases will surpass 35 million, predicted entirely on the growing world population. (DeSantis et al., 2017). Bladder cancer holds the 9th position in the most frequently occurring cancers worldwide. An estimate shows that, in the year 2022, around 614,298 new cases of bladder cancer were reported, causing 220,596 deaths worldwide. Approximately 75% of bladder cancer related burden is experienced by male patients. The rates of mortality and incidence are reportedly highest in Europe, particularly southern Europe (Filho et al., 2026). For the early and accurate detection of bladder cancer, some DNA biomarkers have been established. Among the established biomarkers of bladder cancer, *TERT* promoter mutations are the most prevalent one, occurring in 60-080% cases of all bladder cancer (Cheng et al., 2023). Thus, its detection is important for implementing tailored medicine and precision oncology management. But its detection requires expensive sequencing methods that might not be available in many low-income settings. This study attempts to make a highly sensitive deep learning badsed pre-screening tool to detect *TERT* promoter mutations from conventional histology slides to make the overall process easier and cost-effective.

1.2. About Bladder Cancer

The bladder is a hollow organ located in the lower abdomen and stores urine. Cancer in this organ is a malignancy caused by the uncontrolled growth of abnormal cells in the bladder lining ("Anatomy of the Bladder," n.d.). After prostate cancer, it is the second most common type of cancer of the urinary system ("EAU Guidelines on Prostate Cancer - EPIDEMIOLOGY AND AETIOLOGY - Uroweb," n.d.). It is diagnosed in about 67,200 individuals annually in the United States. Globally, it is the 10th most frequently diagnosed malignancy and stands 13th regarding association with death (Filho et al., 2026) .Bladder cancer is classified into two groups clinically (DeSantis et al., 2017). These are non-muscle invasive bladder cancer (NMIBC) and muscle invasive bladder cancer (MIBC), accounting for approximately 70-75% and 25-30% cases, respectively (Babjuk et al., 2022). While women are inclined to have lower

survival rates, bladder cancer is 3-4 times more common in men than in women (Sharma et al., 2020).

1.2.1. Symptoms of Bladder Cancer

The most common symptom of bladder cancer is hematuria, which occurs in almost 80-90% of patients and stands as the most frequently occurring clinical manifestation (Choo, 2018). It can be immensely visible and results in pink, orange, or dark urine. It can be microscopic as well, which is detected through laboratory testing. Other common symptoms that occur in early cases include:

- Increased urinary frequency.
- Urinary urgency.
- Painful or burning urination, which is described as Dysuria.
- Frequent urination during the night.
- Incomplete bladder emptying.

The common symptoms that occur in the late or advanced stages are:

- Lower back pain, particularly in advanced disease.
- Inability to pass urine.
- Unexplained weight loss and loss of appetite.
- Fatigue and generalized weakness.
- Swelling of the lower extremities and bone pain in metastatic disease. (DeSantis et al., 2017)

1.2.2. Risk Factors of Bladder Cancer

While bladder cancer does not have a definitive cause, there are some habits and environmental factors that might significantly increase the risk of bladder cancer. Tobacco smoking stands as the most significant factor. It accounts for a substantial fraction of cases and increases the risk by approximately two- to four-fold compared with non-smokers (Langone et al., 2012). Occupational exposure to carcinogenic chemicals, such as aromatic amines from the dye, rubber, textile, paint, and hairdressing industries, also increases the risk (Patterson et al., 2026). Gender is another risk factor. Men, mostly older adults, are more susceptible, with nearly 90% of cases occurring in individuals over 55 years of age. Additional risk factors include:

- Drinking and getting exposed to water contaminated with arsenic.

- Chronic bladder irritation due to recurrent urinary tract infections
- Bladder stones
- Infection with *Schistosomiasis* (Jalloh et al., 2020)
- Family or previous personal history of urothelial malignancy
- Genetically predisposing syndromes like Lynch syndrome, mutation in *RB1* and *RAS.*
- Consumption of opium
- Air pollution
- Diabetes mellitus
- Benign prostatic hyperplasia (Grant, 2010).

1.2.3. Types of Bladder Cancer Based on the types of Cells

- **Urothelial Carcinoma (Transitional Cell Carcinoma, TCC):**

  It is the most common histological subtype originating from the urothelial (transitional) cells lining the bladder. It occurs in about 90% of all bladder cancers. It may also occur in the uterus and urethra ("Beyond the bladder," 2022).

- **Squamous Cell Carcinoma (SCC):**

  It is a malignant neoplasm that arises from the squamous cells. It develops within the bladder epithelium due to chronic inflammation. After urothelial carcinoma, it is the second most common histological subtype of bladder cancer and accounts for approximately 3-7% of cases in the United States and other Western countries (Lagwinski et al., 2007). It is also the most common form of non transitional cell carcinoma of the bladder. These types of tumors are mostly present as solitary large masses. Their aggressive nature is reflected by their frequent association with muscular wall invasion during diagnosis.

  Bladder squamous cell carcinoma can be divided into two major subtypes. These are bilharziasis associated squamous cell carcinoma and non-bilharziasis associated squamous cell carcinoma. The former one has a strong association with chronic *Schistosoma haematobium* infection and is commonly observed in areas with this endemic, like parts of Africa, the Middle East, and Asia. The latter one is more commonly observed in Western countries and is typically diagnosed in people in their

70s (“Squamous cell carcinoma of the bladder: pathology, diagnosis and treatment - Shokeir - 2004 - BJU International - Wiley Online Library,” n.d.). It is commonly linked to chronic bladder irritation resulting from long term catheterization, frequent UTIs, and bladder stones (Pottegård et al., 2020).

- **Adenocarcinoma**

  It is a rare histological subtype and accounts for only 1-2% of all bladder cancer cases. It develops from mucus producing epithelial gland like cells, and they are located mostly in the trigone and posterior wall of the bladder. Adenocarcinoma usually presents as a solitary lesion and is invasive during diagnosis. The pathogenesis of this subtype includes chronic irritation to the bladder, which leads to the formation of glandular metaplasia of the urothelium and subsequent malignant transformation (Marks et al., 2023). Papillary adenocarcinoma and signet ring cell adenocarcinoma are histological variants of adenocarcinoma. Overall, bladder adenocarcinoma is associated with poor survival outcomes, with 36.1% and 47.1% reported as 5-year overall and disease free survival rates, respectively (“Types of Bladder Histologies | SEER Training,” n.d.).

- **Small Cell Carcinoma**

  It is a rare, highly aggressive neuroendocrine tumor. It represents only 0.35-0.7% of all bladder cancer cases (Church and Bahl, 2006). Though it's rare, its clinical significance comes from its ability to proliferate rapidly, with its high metastatic potential and poor patient outcomes. Mutations in the tumor suppressor genes like *TP53* and *RB1* are associated with the aggressive biology of this subtype. Patients are often in the advanced stage of the disease and unlikely to have favourable survival outcomes (Ismaili et al., 2009). Median overall survival ranges from about 8.6 to 11.8 months, and the 5-year survival rate is only approximately 14% (Church and Bahl, 2006). Regarding its rarity and aggressiveness, multimodal treatment approaches like chemotherapy, radiotherapy are implemented for its management. However, its outcome remains significantly poorer than the other bladder cancer subtypes (Ismaili et al., 2009).

- **Sarcomatoid carcinoma**

  It is another rare and highly aggressive form of bladder cancer. It accounts for less than 1% of the bladder tumors. It contains both epithelial (urothelial) and mesenchymal (sarcoma-like) cells ("Frontiers | A case series: Four cases of urothelial carcinoma with sarcomatoid differentiation," n.d.).

1.2.4. Classification of Bladder Cancer Based on Tumor Depth

Bladder cancer is broadly classified into two categories based on whether the tumor has invaded the muscle layer of the bladder. These are non-muscle invasive and muscle invasive bladder cancer. Details about them are given below:

- **Non-muscle invasive bladder cancer (NMIBC)**

  Non-muscle invasive bladder cancer accounts for approximately 70-75% of all the bladder tumors. As per its name, this type of tumor is limited to the urothelium layer of the bladder. It is classified into three stages, which are Ta, T1, and carcinoma in situ (CIS/Tis). While, low grade NMIBCs usually do not tend to progress to a more aggressive stage, some high grade T1 and CIS tumors may confer higher risk. Along with that, NMIBC generally has a higher rate of recurrence, and its recurrence has been reported in approximately 80% of cases. It is usually manageable by TURBT and BCG immunotherapy.

  Three subclasses of this type have been revealed based on the mutation and expression signatures through the advancements in high throughput sequencing methods. These are:

  - ➢ Class 1 NMIBC: In this class, a frequent *FGFR3* mutation is observed along with mutations in other genes related to early phases of the cell cycle. Also, expression of *PPARG* and *UPK* is seen to be upregulated.
  - ➢ Class 2 NMIBC: This class exhibits mutations in *TP53* and *ERCC2*. It also demonstrates luminal like differentiation signature.
  - ➢ Class 3 NMIBC: Like class 1, it is also associated with the mutation in FGFR3, but it exhibits basal-like undifferentiated characteristics. ("Molecular Mechanism of SSR128129E, an Extracellularly Acting, Small-Molecule, Allosteric Inhibitor of FGF Receptor Signaling: Cancer Cell," n.d.)

- **Muscle invasive Bladder Cancer (MIBC)**

  As the name suggests, MIBC is defined as its penetration into the detrusor muscle layer or muscularis propia of the bladder wall. It is classified as stage T2 or higher (Holzbeierlein et al., 2024). Approximately 25% of bladder cancer cases at initial diagnosis are of this type. Also, it mostly contains high-grade urothelial carcinomas. In contrast to NMIBC, MIBC is clinically more aggressive and has a higher risk of metastasis to regional or distant organs, generally involving the lymph nodes, lungs, liver, and bones ("Frontiers | Construction of a prognostic signature associated with liver metastases for prognosis and immune response prediction in colorectal cancer," n.d.). Its prognosis highly depends on the stage of the disease. 5-year survival rates were reported to range from approximately 50-70% for the localized cases, while it declined to less than 10% in metastatic cases (Liu et al., 2021).

  The standard treatment for MIBC includes radical cystectomy in association with the complete removal of pelvic lymph nodes. It is often supplemented with perioperative cisplatin-based chemotherapy. Also bladder preserving trimodality, including transurethral resection of the bladder tumor (TURBT), chemotherapy, and radiotherapy, may be considered in some selective cases. Due to its aggressive nature and higher metastatic potential than NMIBC, it requires more attention for its management (Kim and Lee, 2020; Pham and Ballas, 2019).

1.2.5. Grades of Bladder Cancer

Based on how abnormal the cancer cells look under a microscope and how fast they are likely to grow and perform metastasis, the bladder cancer grading system is performed. Two main grading systems are being used worldwide at present. These are the older WHO 1973 three-grade system and the WHO 2004 binary system.

- **WHO 2004 Binary System**

  This is the currently used most standard system that classifies the tumors into two grades. They are high grade and low grade.

  - **Low Grade Bladder Cancer:** It consists of cells that look relatively similar to the normal healthy bladder cells. These cells are well organized and are prone

to grow at a slow pace with a lower tendency to invade the adjacent tissues. The low-grade tumors generally develop as superficial protrusions from the papillae, and they are genetically linked to *FGFR3* mutations ("Molecular characterization of low grade and high grade bladder cancer | PLOS One," n.d.). They consist of approximately 70% of all the bladder cancers. Although they have a high recurrence rate, they show a better prognosis than the high grade version.

- ➢ **High Grade Bladder Cancer**: The high grade tumors contain cells that are poorly differentiated and look unnatural and disorganised under the microscope. They have a propensity to grow rapidly and invade the muscle layer and metastasize. They are genetically linked to TP53 mutations ("Molecular characterization of low grade and high grade bladder cancer | PLOS One," n.d.), and they consist of 30% of all bladder cancer cases. They usually grow from carcinoma in situ (CIS) that directly progresses to become invasive. They are also associated with therapy resistance and worse survival outcomes (Montironi and Lopez-Beltran, 2005).

- **WHO 1973 Three-Tier System:** It is an older method, but it is still widely used as well. It categorizes the tumors into three grades, and these are grade 1, grade 2, and grade 3.
  - ➢ **Grade 1:** It is the low grade version of the 2004 grading system. Cells look pretty much normal and similar to the healthy bladder cells in this case. These are slow growing cells with the least likelihood of spreading and being aggressive.
  - ➢ **Grade 2:** It consists of moderately differentiated tumor cells that look a bit different from the normal bladder cells. They pose moderate risk and tend to grow a bit faster than Grade 1 tumors.
  - ➢ **Grade 3:** They are similar to the high grade tumors containing poorly differentiated cells that look abnormal and disorganized under the microscope. They pose higher risk and are more likely to spread. This grade has the worst survival outcomes among the three grades. (van de Putte et al., 2018).

In the WHO 2004 system, grade 1 and most grade 2 are classified as low grade, and grade 3 and some grade 2 are characterized as high grade tumors.

Table 1.1: World Health Organization grading system of BLCA of the years 1973 and 2004 (Babjuk et al., 2022)

1973 WHO grading system	
Urothelial papilloma	
Grade 1	Well-differentiated
Grade 2	Moderately differentiated
Grade 3	Poorly differentiated
Flat lesions (2004 WHO grading system)	
Urothelial proliferation of uncertain malignant potential (arcinia hyperplasia)	
Reactive atypia (flat lesion with atypia)	
Atypia of unknown significance	
Urothelial dysplasia	

1.2.6. Stages of Bladder Cancer

Bladder cancer is grouped from stage 0 to 4 using the Tumor, Node, Metastasis (TNM) staging system. It describes how deeply the tumor has spread into the bladder wall (Table 1.2).

- **Stage pT0:** When the cancer is non-invasive, it is classified as pT0 stage. It is further divided into two groups: pTa and pTis.
- **Stage pT1:** Here the cancer has invaded the connective tissues but not the muscle layers
- **Stage pT2:** In this stage, the cancer cells start to invade the lymph nodes and other parts of the body outside the bladder. It is also divided into stage 3A and stage 3B. In stage 3A, the cancer has not invaded any distant lymph nodes. It has only grown through the bladder wall into the adjacent tissues. In case of stage 3B, the cancer has invaded lymph nodes near the common iliac artery.
- **Stage pT4:** This is the deadliest among all the stages, and it is divided into stage 4A and stage 4B. In stage 4A, the cancer has spread to the abdominal wall, and in stage 4B, the cancer has invaded other parts of the body like lungs, liver and bones. (“Bladder Cancer Treatment - NCI,” 2023)

Table 1.2: Stages of bladder tumors based on Tumor-Node-Metastasis (TNM) staging system.

T-primary tumor	
T0	No evidence of primary tumor
Ta	Non-invasive papillary carcinoma
Tis	Carcinoma *in situ*: 'flat tumor'
T1	Tumor invades subepithelial connective tissue Tumor invades muscle
T2	T2a Tumor invades superficial muscle (inner half) T2b Tumor invades deep muscle (outer half) Tumor invades perivesical tissue
T3	T3a Microscopically T3b Macroscopically (extravesical mass) Tumor invades any of the following: prostate stroma, seminal vesicles, uterus, vagina, pelvic wall, abdominal wall
T4	T4a Tumor invades prostate stroma, seminal vesicles, uterus or vagina T4b Tumor invades pelvic wall or abdominal wall
N-regional lymph nodes	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in a single lymph node in the true pelvis (hypogastric, obturator, external iliac, or presacral)
N2	Metastasis in multiple regional lymph nodes in the true pelvis (hypogastric, obturator, external iliac, or presacral)
N3	Metastasis in common iliac lymph node(s)
M-distant metastasis	
M0	No distant metastasis M1a Non-regional lymph nodes M1b Other distant metastases

Here, the stages pT0 to pT1 are for the non-muscle invasive bladder cancer (NMIBC) cases, and the stages pT2, pT3, and pT4 fall into the category of muscle invasive bladder cancer (MIBC).

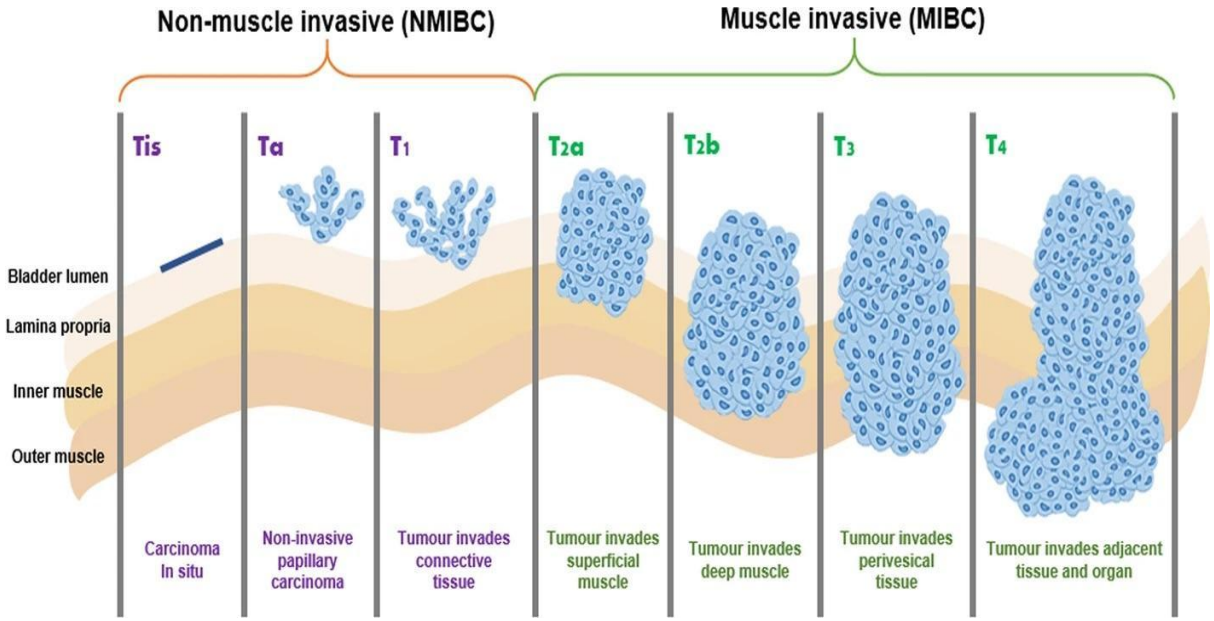


Figure 1.1: Different stages of bladder cancer. Tis, Ta, and T1 fall into the non-muscle invasive (NMIBC) category. T2a, T2b, T3, and T4 fall into the MIBC category (Kanmalar et al., 2022).

1.2.7. Treatments of Bladder cancer

- **Transurethral Resection of Bladder Tumor (TURBT):** This is the primary treatment for NMIBC where the tumor is surgically removed through the urethra by a cystoscope. Often electrical current or laser is applied along with the procedure to destroy the remaining cancer cells.
- **Intravesical Therapy:** It includes the use of Bacillus Calmette-Guerin (BCG) vaccine or chemotherapy. After TURBT, use of intravesical BCG immunotherapy is the standard treatment for high-risk NMIBC cases. It effectively reduces the recurrence risk and lowers the poor survival outcome rates. BCG is a live attenuated vaccine, and it is placed directly into the bladder to trigger the cancer surveillance mechanism of the immune system.

  For cases that are BCG-unresponsive, the treatment includes the use of other intravesical agents like Valrubicin (chemotherapy), and pembrolizumab (immunotherapy) ("New Treatment Options for Non–Muscle-Invasive Bladder Cancer | American Society of Clinical Oncology Educational Book," n.d.)
- **Neoadjuvant Cisplatin-Based Chemotherapy:** It is the standard care used before the surgical resection of bladder tumor in MIBC cases. It is associated with about a 5%

increase in survival rate at 5 years and a 14% reduction in relative risk for death. Common chemotherapy combinations are:

- ➢ Gemcitabine + Cisplatin (GC)
- ➢ Methotrexate, Vinblastine, Doxorubicin, Cisplatin
- ➢ Gemcitabine + Paclitaxel (Roviello et al., 2022).

- **Radical Cystectomy:** It is the standard surgical treatment for MIBC cases. It involves the complete bladder removal with dissection of pelvic lymph nodes. In men, it also involves the removal of the prostate, and in women, it involves the removal of the uterus and part of the vaginal wall (Milowsky et al., 2016).
- **Trimodality Therapy:** It is the bladder-preserving therapy. It is performed for patients who can not undergo cystectomy. It is used as an alternative to surgery. Here, a combination of TURBT, radiation therapy, and chemotherapy is employed to treat the cancer.
- **Immune Checkpoint Inhibitors:** It is another sort of immunotherapy. It is used as second-line therapy after platinum chemotherapy. The FDA-approved immune checkpoint inhibitors are:
  - ➢ Pembrolizumab
  - ➢ Atezolizum,ab
  - ➢ Nivolumab
  - ➢ Avelumab (Lopez-Beltran et al., 2021)
- **Antibody-Drug Conjugates:** Enfortumab vedotin targets Nectin-4. It is used for metastatic urothelial carcinoma after platinum chemotherapy.
- **Targeted Therapy:** Erdafitinib is an FGFR inhibitor used for patients with *FGFR3* mutation.

1.2.8. Diagnosis of Bladder Cancer

Bladder cancer diagnosis includes a mixture of clinical evaluation, urine tests, imaging, and tissue biopsy.

- **Urinalysis:** It is used to detect hematuria or the presence of microscopic blood in the urine. Hematuria is the primary possible indicator of bladder cancer.
- **Urine Cytology:** Here, urine is examined under a microscope to identify cancer cells. It has low sensitivity but high specificity. It is more effective for high grade cases than the low grade ones (Planz et al., 2005)
- **Cystoscopy:** It is the gold standard for the initial diagnosis of bladder cancer. In this case, a thin camera is inserted through the urethra to visualise the bladder lining directly. However, the procedure comes with several complications, including UTI, inability to urinate, and severe injury ("Cystoscopy & Ureteroscopy - NIDDK," n.d.)
- **Transurethral Resection of Bladder Tumor (TURBT):** During cystoscopy, surgeons can take out the tumor and perform biopsy to look for the presence of cancer cells, and diagnose stages and grades (Magers et al., 2019).
- **Urine-Based Biomarkers:** These are non-invasive diagnostic approaches to detect cancer cells, proteins, DNA mutations, or methylation patterns shed from the bladder into the urine. It offers a painless alternative to cystoscopy. Prominent markers include DNA markers (TP53, TERT, FGFR3), protein markers (NMP22, BTA), methylation markers (FAM19A4/GHSR/MAL) (Maas et al., 2023a).
- **Biomarkers based on DNA mutation:** these are based on point mutations that are frequently observed in bladder cancer. One of the most common examples of this is the mutation in the telomerase reverse transcriptase (*TERT*) gene promoter. In this region, C228T and C250T are the most prevalently occurring mutations (Flores Monar et al., 2023). Additionally, *HRAS, TP53,* and *FGFR3* are also often mutated in bladder cancer ("UroSEEK gene panel for bladder cancer surveillance - Wong - Translational Andrology and Urology," n.d.). Methods like ddPCR and NGS have been developed to detect these mutations.

1.3. About Telomerase Reverse Transcriptase (*TERT*) Gene

TERT gene is located on the short arm of chromosome 5. It is positioned very close to the 5p telomere end. Its position is important for its regulation ("Regulation of the Human Telomerase Gene TERT by Telomere Position Effect—Over Long Distances (TPE-OLD)," 2016). It encodes the catalytic protein component of telomerase. Telomerase is an enzyme that uses its

reverse-transcriptase activity to maintain the length of telomeres, the protective ends of chromosomes (Maas et al., 2023b).

1.3.1. Physiological Role of *TERT*

TERT encodes the catalytic subunit of telomerase. Telomerase functions by adding repeated sequences of DNA (TTAGG) to the 3' ends of the chromosomes (Figure 1.2). *TERT* protein binds to the telomerase RNA component, which serves as the template. Then, telomerase catalyses reverse transcription. The entire cycle involves primer binding, extension, and release (Maas et al., 2023b).

This process has a wide range of biological effects. It is involved in cellular replication and senescence. It prevents the gradual shortening of telomeres that occurs during normal cell division. It is also involved in chromosomal repair through de novo synthesis of telomere repeats at double-stranded DNA breaks. It also plays an important role in aging and preventing programmed cell death. It is primarily more active in progenitor cells, stem cells, germ cells, and cancer cells, and repressed in normal adult cells. It is often reactivated during oncogenesis (Yuan and Xu, 2019; (Maas et al., 2023b).

1.3.2. *TERT* Promoter Mutations & their Molecular Impact

Two major somatic point mutations occurring in this region are C228T and C250T, accounting for 57.7% and 15.5% of mutations, respectively. Mutations in the *TERT* promoter result in a 2-5-fold increase in *TERT* mRNA and protein expression, increased telomerase activity, prevention of replicative senescence, and cellular immortalization. Together, these events promote tumor growth and evasion (Li et al., 2015; El Azzouzi et al., 2024).

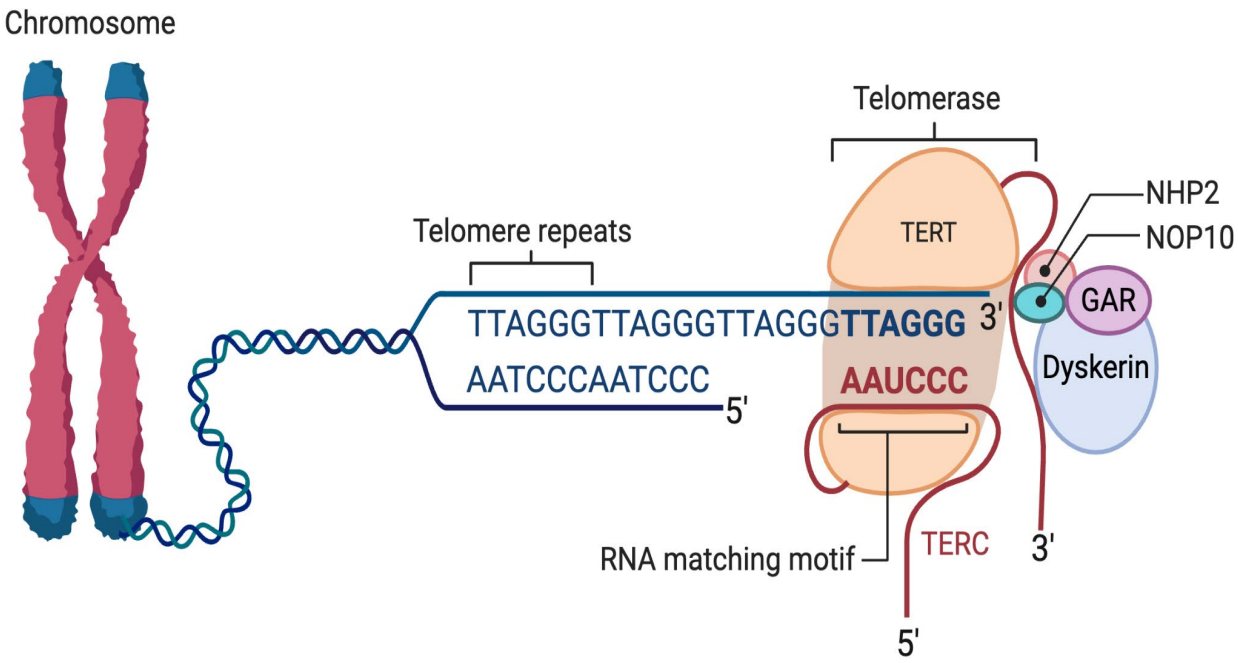


Figure 1.2: Molecular mechanism of *TERT* protein in normal telomerase function. *TERT* protein acts as a catalytic subunit of telomerase and performs reverse transcription to prevent shortening of telomeres (Marchese et al., 2022).

C228T Mutation: It is located at -124 bp upstream of the transcriptional start site. It creates a de novo 11bp binding motif (5'-CCCCTTCCGGG-3') containing the GGAA consensus sequence for the ETS family of transcription factors (Tang et al., 2013). GABPA, ETS1, and ETS2 are part of this family of transcription factors. The de novo ETS binding site is located at an ideal distance from the native ETS binding site. This positioning allows GABP tetramer formation, and it binds and activates the promoter (Figure 1.3). It alone is sufficient for the activation of the promoter and leads to an increase of 2-4-fold transcriptional activity without necessitating other signaling mechanisms. It results in the production of 2-5-fold higher *TERT* mRNA and protein, which ultimately leads to cellular immortalization ("The transcription factor GABP selectively binds and activates the mutant TERT promoter in cancer | Science," n.d.)

C250T Mutation: It is located at -146bp upstream of the transcriptional start site. It also creates the same 11bp de novo ETS binding site as the C228T mutational subtype with the exception of requiring cooperative binding with non-canonical NF-κB signaling for *TERT* activation (Y. Li et al., 2015a). The NF-κB half site is located immediately adjacent to the mutated ETS motif. It allows p52 and ETS1/ETS2 to bind cooperatively to the same promoter region (Figure 1.3).

This dual requirement of both ETS and NF-κB makes C250T activation more conditional (Najarro et al., 2015; Raihan et al., 2018).

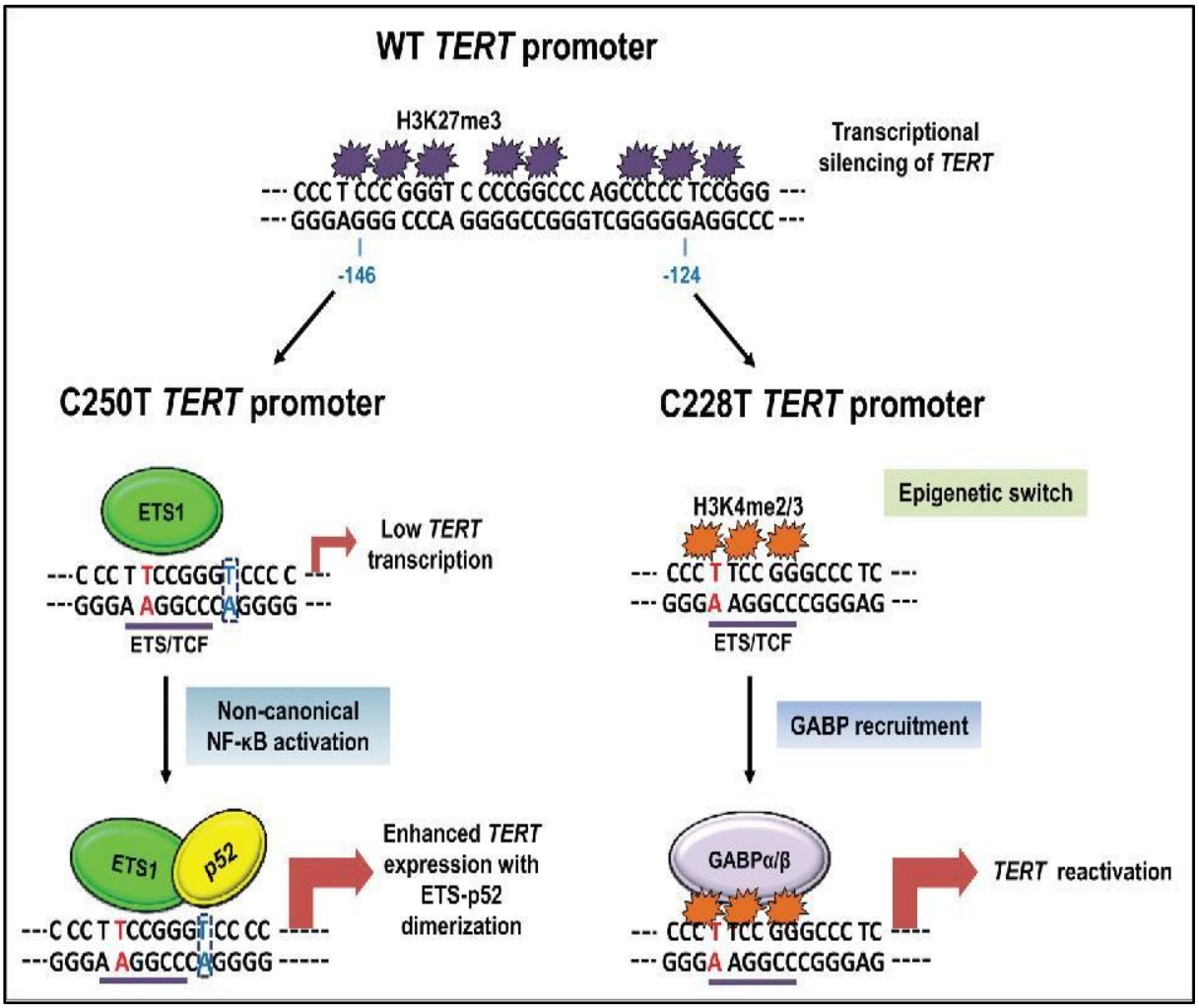


Figure 1.3: Molecular Impact of *TERT* promoter mutations. Mutation at both subtypes, C228T and C250T, generates the de novo 11bp ETS bonding site. Mutation at C250T functions through a non-canonical NF-κB activation, while mutation at C228T functions through GABP recruitment. Both mutations result in overexpression of *TERT* mRNA and protein (Li and Tergaonkar, 2016).

1.3.3. Frequency of *TERT* Promoter Mutation in Bladder Cancer

This is the most prominent mutation in bladder cancer. It occurs in about 60-80% of patients globally. C228T has a frequency of 57.7%, while C250T is less common, consisting of only 15.5% of total mutations. Approximately 78.9% frequency of the *TERT* promoter mutation has been reported in large scale studies. This mutation occurs independently across all grades and stages of urothelial bladder cancer. Its frequency is observed at about 74% and 66% in non-muscle invasive precursor lesions and muscle invasive bladder cancer, respectively. Its high frequency in NMIBC indicates that these mutations occur as an early event during oncogenesis. *TERT* mutations are associated with *FGFR3* mutations ($p = 0.0002$) and are detected in 60.4% of NMIBC cases, while their prevalence is 23.5% in MIBC tumors. This phenomenon suggests the higher prevalence of cooccuring these both mutations in less aggressive cases. Also, these mutations are detectable in urine up to 10 years before clinical diagnosis, making them an ideal marker for liquid biopsy in early detection of bladder cancer (Jalkanen et al., 2023).

1.4.1. Use of Artificial Intelligence and Deep Learning to Detect Mutations from Medical Images

Incorporation of artificial intelligence (AI) and deep learning in the analysis of regular medical images provides an alternate way of mutation detection that is rapid, cost-effective, and non-invasive. Deep learning algorithms can identify mutations from conventional histological slide images of cancers in 13 out of 14 tumor types, such as lung, gastric, breast, and colorectal cancers (“Deep learning algorithm identifies tumor subtypes based on routine histological images - UChicago Medicine,” n.d.).

Detection of mutations in genes like *EGFR, KRAS, BRAF,* and categorization of histological subtypes are also possible using convolutional neural networks (CNN) with AUC 0.88-0.95 (Ishtiaq and Farouk, 2025). Identification of survival prognosis, treatment efficacy, and recurrence are also possible along with the mutation detection by incorporating AI into radiology image analysis (Chakrabarty and Mahajan, 2024). It is reported that breast cancer detection is improved using deep learning methods, and it enables the prediction of long term risk as well (“AI and Cancer - NCI,” 2024).

Inclusion of multiple instance learning (MIL) can reportedly predict certain gene mutations from whole slide images in bladder cancer. This process utilizes masking (separation of part of cancerous tissue from the whole slide), patch extraction, and detection of mutation(Yan et al.,

2023). Deep learning can identify mutations in *FGFR, FGFR2, FGFR3, ERBP2, TP53, PI3K* from histological images of bladder cancer with reported AUCs of (0.76-0.94) ("Using deep learning to identify bladder cancers with FGFR-activating mutations from histology images - Velmahos - 2021 - Cancer Medicine - Wiley Online Library," n.d.) (Yan et al., 2023). This process can reduce the sequencing burden and cost in resource limited settings.

1.4.2. About Phikon-V2

Phikon-V2 is a vision transformer. It is self supervised and designed particularly for histopathological image analysis. It is developed by Owkin. For its development, 460 million tiles from pathological images from 58000 WSIs across 30 cancers were used for its training. The images used for the training were captured at 20x magnification. It is a lightweight vision transformer model and is publicly available (Filiot et al., 2024). Phikon-V2 is trained on a large number of histopathological images from diverse cancer types. This high scale and diversity make it unique from other deep learning based feature extractors. It uses DINOv2 for its self-supervised learning without providing any sort of manual labels ("LightlyTrain Benchmarks for Histology: SSL Pretraining and Segmentation," n.d.). Phikon-V2 surpasses the original Phikon model in terms of its performance since the original version was only pre-trained on 40 million histological tiles (Filiot et al., 2024).

Phikon-V2 works by taking patches of histological images captured at 20x magnification. Then it passes the tiles through the ViT-L architecture, and DINOv2-mediated self-supervised pre-training is employed. It results in the feature vector in CLS token form as the output. It has been used in biomarker discovery, mutation identification from histology images, molecular pathway prediction, survival analysis, etc (Filiot et al., 2024).

1.5. Objectives of the Study

- Identification of *TERT* promoter mutation in C228T and C250T subtypes in our bladder cancer cohort.
- Determination of *TERT* promoter mutation frequency in bladder cancer in our population and comparing it with other published studies
- Generation of a deep learning based pre-screening tool that can identify *TERT* promoter mutations from routine histological slides of bladder cancer with high sensitivity, while accounting for grade related confounding, to reduce cost and burden of expensive sequencing methods in resource limited settings.

Section 2

Methods and Materials

2.1. Overall Experimental Framework

The study involves the collection of 77 bladder cancer tumor samples and the generation of their histological images and evaluation of their histological grade. Also, it incorporates the extraction of DNA from the tumor samples to identify the p*TERT* mutation using ddPCR, followed by using that information as ground truth for the subsequent development of a deep-learning model, which will act as a pre-screening tool to detect p*TERT* mutational status from conventional H&E-stained histological images to reduce the sequencing burden in laboratories operating in resource-limited settings (Figure 2.1).

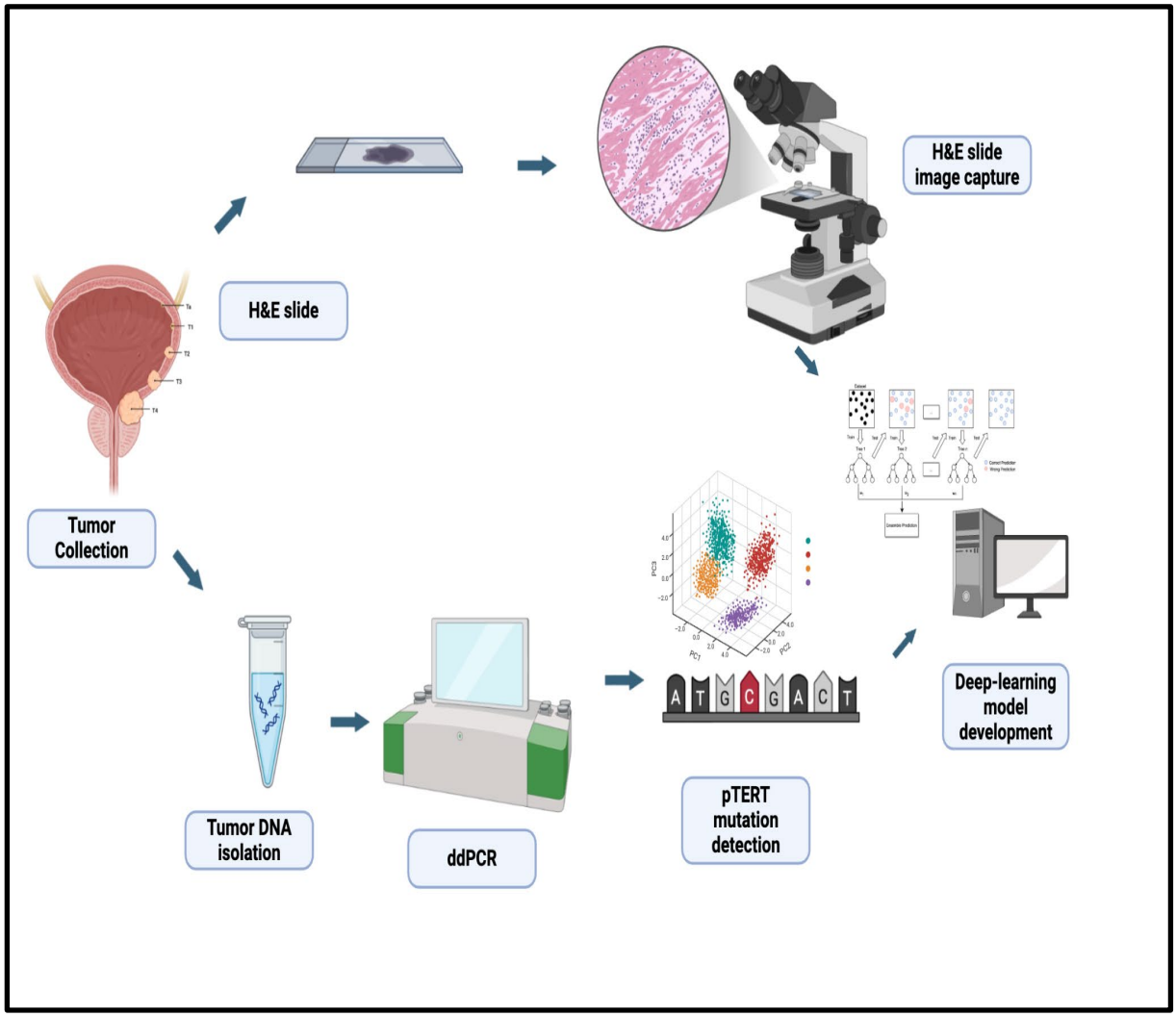


Figure 2.1: An overview of the entire experiment. It describes the bladder cancer tumor sample collection and subsequent analysis; one arm involves the generation of histopathological images and another arm depicts the generation of ground truth using ddPCR and making a deep learning model using both information.

2.2. Ethical Approval

Bangladesh Medical Research Council approved the study. All patients gave written informed consent, which stated that the data collected would be used exclusively for research purposes. All relevant documentation, including ethical clearance certificates and consent forms, has been included in the Appendix section.

2.3. Collection of Tumor Tissue and Its Processing

Tumor samples were acquired only from the patients who had signed the written consent form. Following surgical resection, the samples were immediately transferred into RNAlater-containing tubes by the operating surgeon. The samples were kept on ice during transportation, and subsequently stored at -80°C. Clinical information involving age, sex, and previous medical history was obtained from the questionnaires. These were then validated against previous medical records. To reduce bias, tumor histology was evaluated in an independently blinded manner to identify the grade and stage of cancer.

2.4. Collection of Clinical Metadata through Questionnaires

All the participants provided their information on age, height, sex, and medical history in the questionnaires. The process was performed with the aid of a trained clinician with the full consent of the patients.

2.5.1. DNA Extraction

DNA was extracted from the bladder cancer tumor tissue by using the DNeasy Blood & Tissue Kit from Qiagen (Figure 2.2). Though the kit included all necessary reagents and extraction manuals, the protocol was optimized to achieve the maximum yield of DNA from the tissue samples. Tissue samples were transferred from 80°C to -20°C the day before extraction, then they were thawed on ice on the day of the procedure. Following that, approximately 30 mg of tissue was transferred into a sterilized mortar and pestle. Liquid nitrogen was used to facilitate the grinding process and to produce fine powder.

900 µL of lysis buffer was added to the powdered tissue, and they were gently mixed. After that, the mixture was transferred into a sterile Eppendorf tube, and 4.5 µL of Proteinase K was added to it. The mixture was then incubated at 55°C for three hours and kept at 4°C overnight afterwards. The next day, 4.5 µL RNase inhibitor was added, and the mixture was again

incubated at 37°C for one hour, followed by brief cooling on ice for three to five minutes. 300 µL of protein precipitation solution was added and vortexed for 20 seconds. The precipitated proteins were then removed by centrifugation at 16000 RCF for three minutes. The DNA-containing supernatant was carefully transferred to a fresh tube.

900 µL of ice-cold isopropanol was introduced to precipitate the DNA, and it was pelleted again by centrifuging at 16000 RCF for two minutes. The pellet was then washed using ethanol and resuspended in 200 µL of DNA hydration solution by using a brief vortex, followed by incubation at 65°C for one hour. After that, the concentration and purity of DNA were evaluated using a NanoDrop Spectrophotometer, and the extracted DNA was stored at -80°C for subsequent use.

2.5.2. Assessment of extracted DNA quality

The concentration of extracted DNA was measured using the Thermo Scientific NanoDrop One/Onec Microvolume UV-Vis Spectrophotometer. It concurrently assessed DNA purity by measuring two absorbance ratios. These are:

1. A 260/280 ratio was used to identify potential RNA and protein contamination. A value of 1.8 served as the standard for pure DNA.
2. On the other hand, a 260/230 ratio indicated contamination with phenol, salts, or carbohydrates. Values exceeding 2.0 were considered acceptable.

Any samples failing to meet these purity thresholds were subjected to re-extraction to ensure only high-quality DNA was used in downstream applications. Before PCR, all extracted DNA samples were diluted to a working concentration of 20 ng/µL.

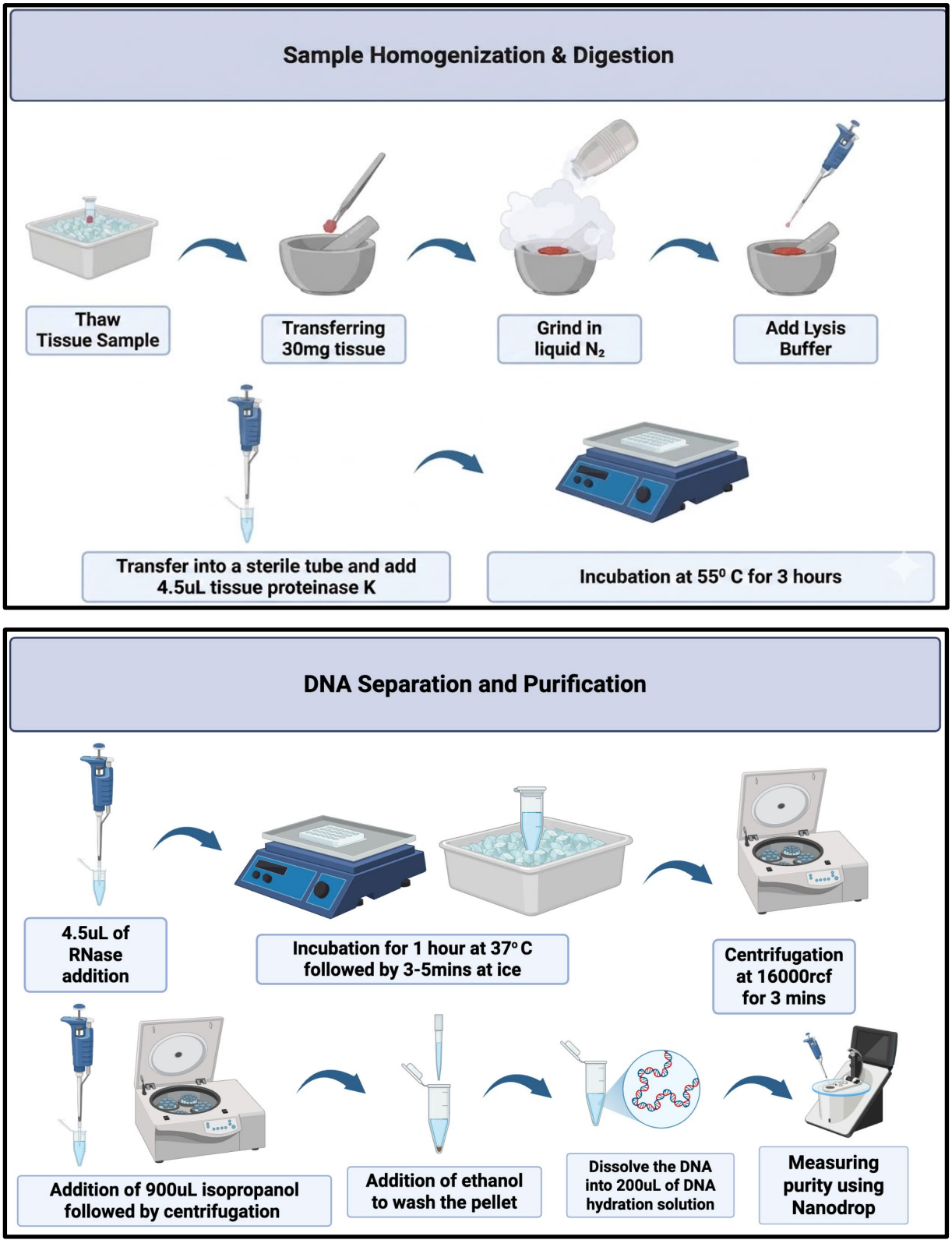


Figure 2.2: A comprehensive workflow of DNA extraction from tumor tissue samples. The top segment shows detailed procedures of sample homogenisation, digestion with proteinase K, and finally incubation. The bottom segment depicts the separation of DNA from the mixture by using RNase and purification of the DNA.

2.6. Droplet Digital PCR (ddPCR)

A ddPCR assay was developed to detect the *TERT* promoter mutation in two hotspots for bladder cancer tumor DNA (Figure 2.3).

2.6.1. Primer and Probe Design

The probes and primers were made using Primer3Plus, employing genome coordinates from the study by Song Wu et al., after converting them to hg38 from hg19. Three allele-specific probes tagged with different fluorophores were designed to distinguish the wild-type allele from the two *TERT* targeted promoter mutations. The wild-type probe was labeled with HEX, and the mutation-specific probe against the C250T mutation (chr5:1,295,135 (C>T)) was tagged with FAM, and the probe targeting the C228T mutation (chr5:1,295,113 (C>T)) was labeled with FAM (Table 2.1). NCBI Primer-BLAST was used to evaluate the specificity of the primers before experimental application. Forward and reverse primers were subsequently elongated using the QX600 Droplet Digital PCR System by Bio-Rad to generate ddPCR data.

Table 2.1: List of dyes used to visualise wild-type alleles and different mutational subtype-containing alleles.

Probe Target Coordinate	Dye
Wild-Type	HEX
chr5:1,295,135 (C>T)	FAM
chr5:1,295,113 (C>T)	FAM

2.6.2. Optimization for Target Amplification Conditions

Gradient PCR was used to establish the optimal annealing temperature for both primers and probes. Plasmid constructs containing target sequences were used to optimize mutation detection. Different ratios of wild-type and mutant alleles were assessed to ascertain the minimum detectable threshold.

2.6.3. Detailed Methodology

The DNA template from each sample was mixed with Bio-Rad master mix, allele-specific probes, and primers to prepare for the reaction. After that, 20 μL of the reaction mixture was transferred into a droplet generation cartridge. Then, the droplet generation oil was incorporated into the designated channel. The cartridge was then placed in the QX100 Droplet Generator, where microfluidic processing produced an emulsion of approximately 20,000 nanoliter-sized droplets per sample. The resulting emulsion was later loaded into PCR tubes for amplification.

The droplet emulsion was subjected to thermal cycling employing standard PCR conditions using a thermal cycler for PCR amplification. Taqman probes were hybridized to their corresponding targets and subsequently cleaved during amplification. This process generated signals. After the amplification process, droplets were analysed using the QX600 Droplet Reader. Each sample was aspirated using the autosampler, which separated individual droplets and directed them through a fluorescence detector for signal detection.

2.6.4. Analysis of ddPCR Data

QX Manager Software was employed for ddPCR data analysis. Fluorescence data were visualized in 1D, 2D, and 3D plots based on signal intensity and emission profiles. According to the fluorescence characteristics, droplets formed distinct clusters, and each cluster represented a specific allele. Cluster identities were annotated automatically by the software based on predefined thresholds. The results were further evaluated manually to confirm the accuracy and label the genotype of each sample. Genotype information from ddPCR was used as a ground truth in the downstream deep-learning model development.

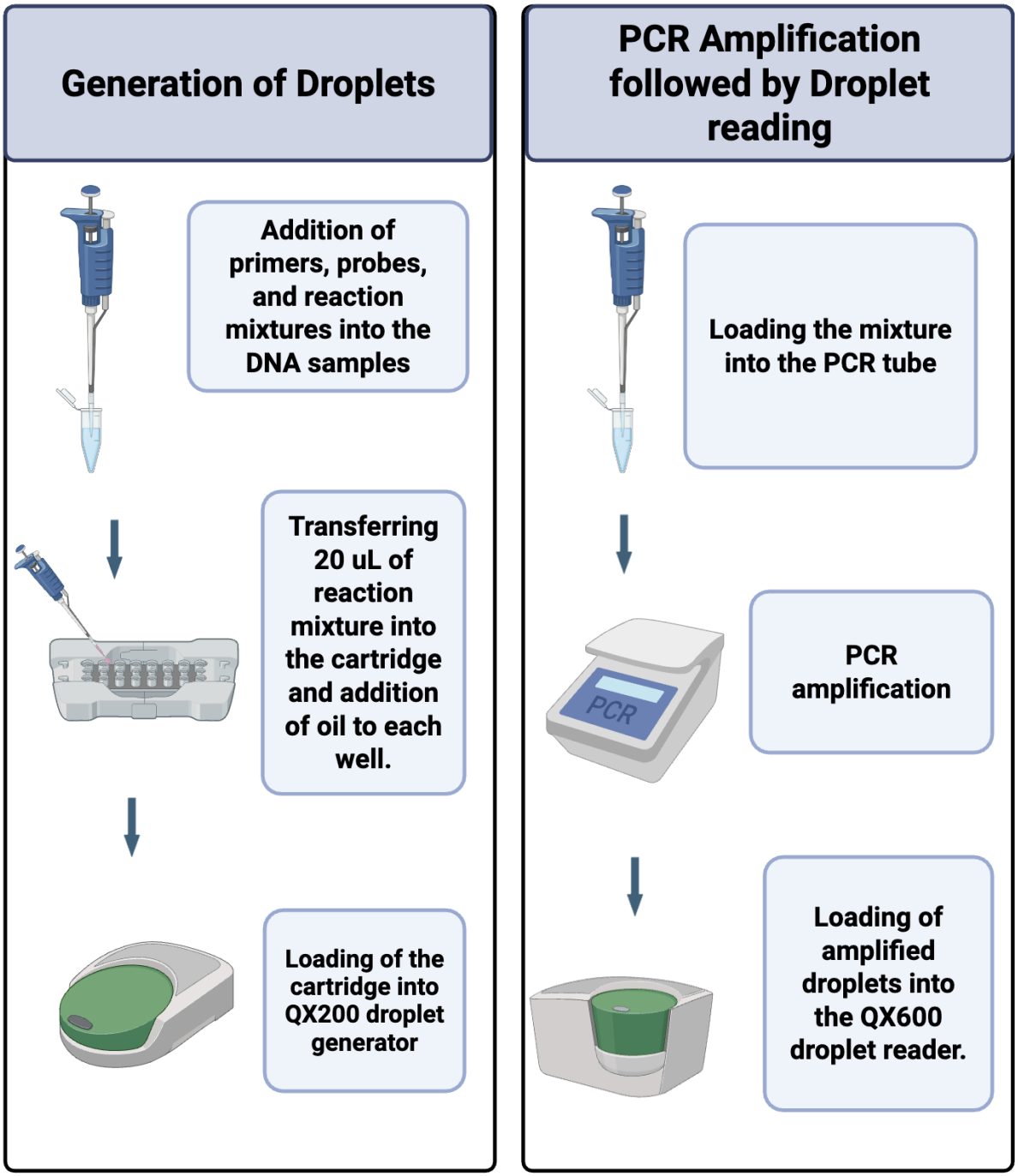
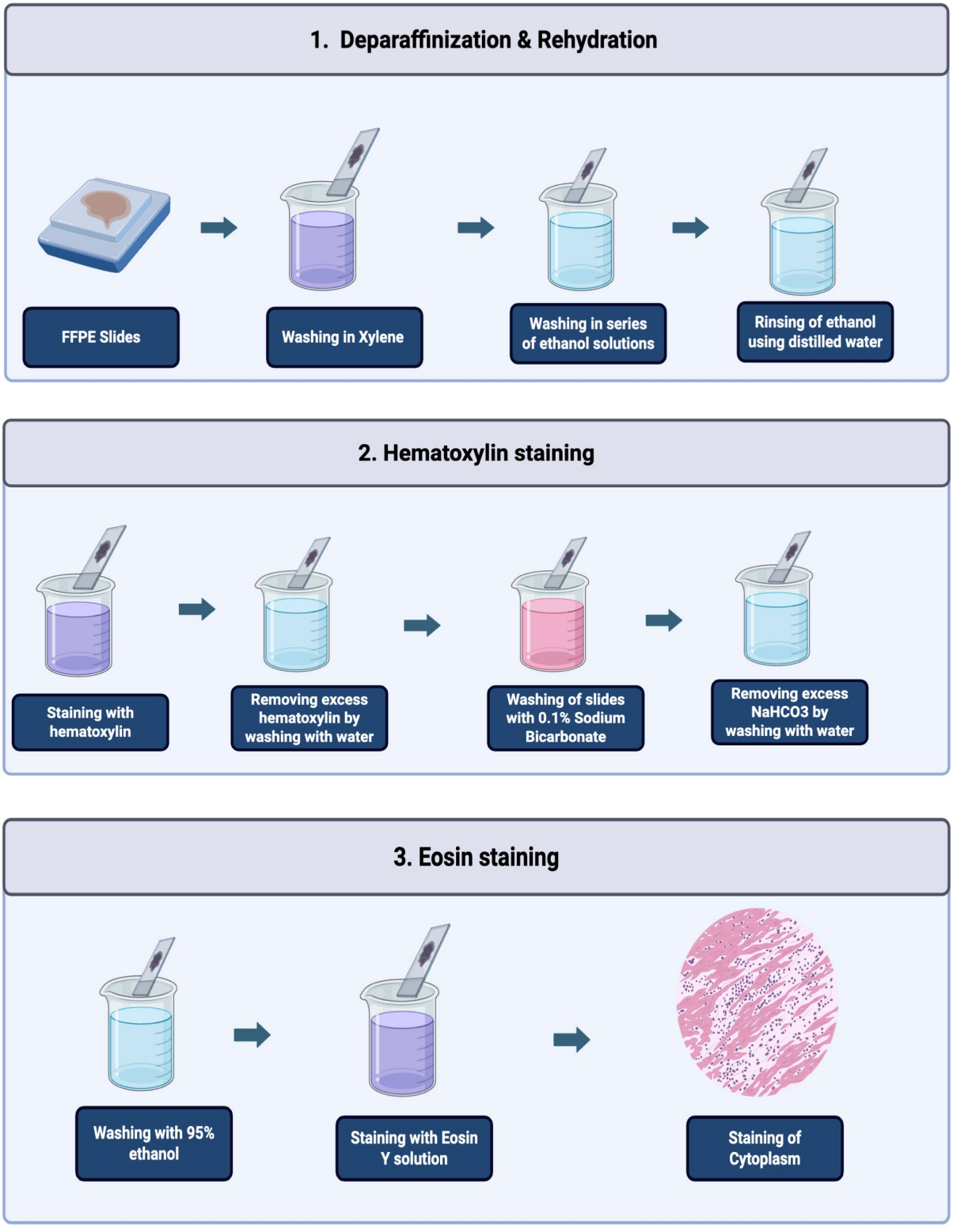


Figure 2.3: An overview of the droplet digital PCR technique. It involves the sequential steps that result in the generation of droplets, which are further subjected to PCR amplification, followed by droplet reading to identify the target mutation.

2.7. Hematoxylin and Eosin (H&E) staining

1. **Deparaffinization and Rehydration:** FFPE blocks were provided by the pathology department. 4 μm thick FFPE tissue sections were cut using a microtome, and they were mounted onto glass slides. To ensure adequate tissue adhesion, the sections were dried overnight at 37 °C. The slides were immersed in xylene for 5 minutes for two consecutive washes to deparaffinize them. After that, the tissue sections were gradually rehydrated by washing them with a series of gradually decreasing strengths of alcohol solutions. First, they were treated with 100% ethanol for 5 minutes, then they were placed in 95% ethanol for 5 minutes, followed by 70% ethanol for 5 minutes, and finally 50% ethanol for 5 minutes. Then, distilled water was used for 2 minutes to rinse the slides and remove leftover alcohol from them.
2. **Hematoxylin Staining:** The rehydrated sections were then immersed in Mayer's hematoxylin solution for 5 minutes to stain the nuclei. Then the excess hematoxylin was removed by placing the slides in the running tap water for 5 minutes. After that, the slides were dipped into 0.1% sodium bicarbonate solution for 1-2 minutes, which shifted the hematoxylin color from red to blue-purple. Finally, another round of washing using tap water was performed to remove the excess sodium bicarbonate.
3. **Eosin Staining:** The sections were dipped in 95% ethanol for 30 seconds, followed by immersion into eosin Y solution (0.5-1% eosin in 95% ethanol) for 1-3 minutes. It stained the cytoplasm and extracellular matrix in pink colour.
4. **Dehydration and Mounting:** After eosin staining, the slides were dehydrated again by washing them with a series of gradually increasing strengths of alcohol solutions. First, they were treated with 95% ethanol for 2 minutes, then they were washed twice with 100% ethanol for 2 minutes. The sections were then rehydrated by washing them twice with xylene for 2 minutes. After that, the slides were mounted using DPX and coverslips. Then, these were left to dry in a fume hood overnight before microscopic examination.
5. **Microscopic Evaluation:** An Olympus BX53 microscope was used to examine the tissue sections. Nuclei appeared dark blue or purple, while the stroma appeared pink to red. The stained sections were evaluated by a pathologist to confirm the histological diagnosis and identification of tumor grade and stage of the bladder cancer tissue (Figure 2.4).


1. Deparaffinization & Rehydration
FFPE Slides
Washing in Xylene
Washing in series of ethanol solutions
Rinsing of ethanol using distilled water
2. Hematoxylin staining
Staining with hematoxylin
Removing excess hematoxylin by washing with water
Washing of slides with 0.1% Sodium Bicarbonate
Removing excess NaHCO3 by washing with water
3. Eosin staining
Washing with 95% ethanol
Staining with Eosin Y solution
Staining of Cytoplasm

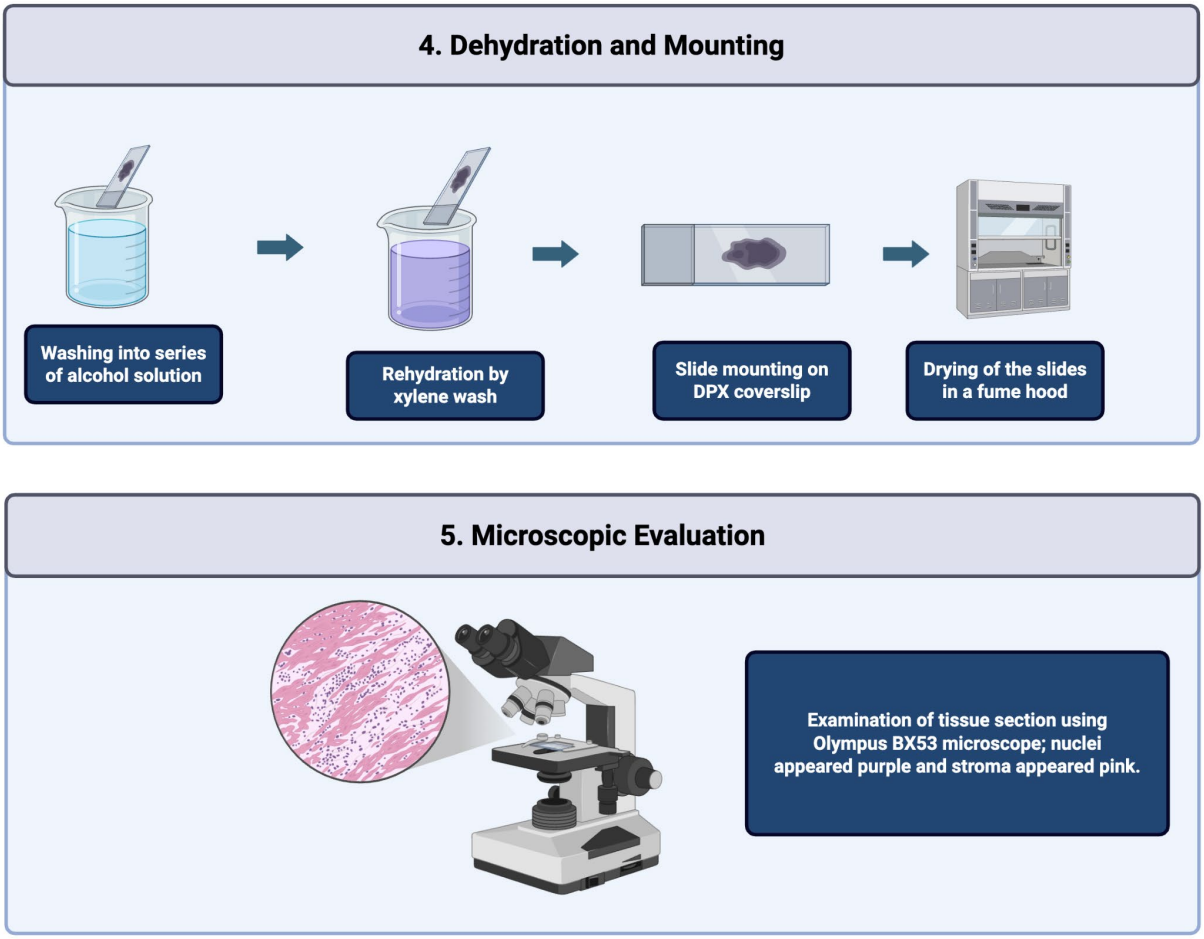


Figure 2.4: A detailed representation of the hematoxylin and eosin staining procedure of tumor tissue. 1. Depicts the deparaffinization and rehydration of the FFPE slides using xylene and ethanol solution. 2. It shows the proper staining of tissues using hematoxylin, with removal of excess staining using sodium bicarbonate. 3. It depicts the staining of stroma resulting from the eosin staining of the tissue. 4. It visualises the dehydration of the slides in a fume hood after their mounting on a DPX coverslip. 5. Representation of how the H&E-stained slides look under a microscope; the nuclei appear bluish purple, while the stroma appear pink.

2.8. Development of the Pre-screening Deep Learning Model

All computational analyses were conducted on Python 3.12 using Google Colaboratory.

2.8.1. Preprocessing of H&E-stained Histology Images and their Stain Normalisation

Overall, 105 H&E-stained cancer tissue images were extracted from Google Drive, which correspond to 77 unique patients. Images were stored in TIFF format at a resolution of 1920x1440 pixels. Images were captured using an Olympus DP22 digital camera at 20x magnification, leading to an approximate spatial resolution of 0.25 µm/pixel.

Preceding feature extraction, Mackenko stain normalisation was applied to all images to diminish inter-slide colour variation originating from batch and processing differences

("Macenko Method for Normalizing Histology Slides for Quantitative Analysis - GeeksforGeeks," n.d.). The first image in the dataset (62.tif) was selected as the reference slide. For each image, pixel intensities were converted to optical density (OD) space using the Beer-Lambert Law using the numy package (Harris et al., 2020). Then, the mean and standard deviation for all three OD values were calculated for the reference image throughout all three colour channels. All the images were normalized by standardising their OD distributions to match the reference statistics. Then, the normalized images were converted back into the RGB colour system before feature extraction using the PIL package ("Pillow," n.d.). For images that couldn't be normalised due to poor tissue content, their original version was retained as a backup.

2.8.2. Feature Extraction using Phikon-v2

Phikon-v2 was used to extract the deep morphological features from each normalized image. It is a vision transformer-based model developed by Owkin. The transformer was loaded from the Hugging Face model repository (owkin/phikon-v2) through the AutoModel and AutoImageProcessor interfaces from the Transformer library. The model was only used as a feature extractor, and its weights were kept frozen (Filiot et al., 2024).

The normalised RGB images were preprocessed again by the phikon-v2 image processor. It resized (converted the images into 224 x 224 pixels) and normalized the images according to the pretraining parameters of the model. Then, these preprocessed images were passed through the vision transformer encoder. After that, from the final transformer layer, the output [CLS] token of each image was extracted as a 1024-dimensional feature vector using the NumPy package. These tokens represented the global morphological features of the tissue section (Filiot et al., 2024; Harris et al., 2020).

2.8.3. Phikon-v2 Extracted Feature Analysis

Three hand-crafted proxy features were estimated to gain additional explanatory insights into the nature of the features extracted by Phikon-v2 using the scipy.stats package (Filiot et al., 2024; “Statistical functions (scipy.stats) — SciPy v1.18.0 Manual,” n.d.). These are:

- Intensity proxy: it represents the pixel brightness across all channels and reflects overall staining density.
- Texture proxy: it represents variance, which reflects tissue heterogeneity.
- Edge proxy: it represents the mean gradient value calculated from the grayscale image. It reflects the nuclear and structural boundary sharpness.

Spearman's rank correlation was estimated between the 1024 Phikon-V2 features and the three proxy features. Then, each feature was assigned to its representative proxy feature or the one with which it showed the highest correlation (Table 2.2).

Table 2.2: The types of features extracted using Phikon-V2 from the histological slide images of bladder cancer.

Feature Type	Count	Percentage
Edge	508	49.61%
Intensity	267	26.07%
Texture	249	24.32%

2.8.4. Patient Level Aggregation

For patients with more than one image, features were first extracted independently from each image, and then they were averaged to produce a single representative 1024-dimensional feature vector per patient. This ultimately resulted in a matrix of shape (77 patients x 1024 features). *TERT* promoter mutation labels from simplex ddPCR of tumor tissue (mutant = 1, wild type = 0) and Grade (High = 1, Low = 0) were aligned to each patient appropriately. It was done using NumPy and the Pandas package (Filiot et al., 2024; Harris et al., 2020; “pandas - Python Data Analysis Library,” n.d.).

2.8.5. Classifier Selection

A systematic comparison was applied across multiple algorithms and hyperparameter configurations using Leave-One-Out cross-validation (LOO-CV) to identify the optimal classifier using sklearn.ensemble and sklearn.model_selection package ("sklearn.ensemble," n.d.; "sklearn.model_selection," n.d.). The following classifiers were computed:

- Logistic Regression with C= 0.1, 1.0, 5.0, 10.0 as a regularization parameter.
- Gradient Boosting Classifier with different tree depths (d= 2, d= 3) and learning rates ("GradientBoostingClassifier," n.d.).
- Support Vector Machine with RBF and linear kernels.

For each classifier, feature counts of 20, 30, 45, 60, and 100 were assessed. Features were standardized using StandardScaler within each LOO-CV. ANOVA F-statistics using the f_classif package were applied to detect the top-N features most correlated with *TERT* promoter mutation status, and this was strictly done on the training data to avoid data leakage ("f_classif," n.d.).

The Gradient Boosting Classifier with 50 estimators, depth 3, and learning rate 0.1 consistently achieved higher AUC than other classifiers with 100 features ("GradientBoostingClassifier," n.d.).

2.8.6. Hyperparameter Optimisation

After selecting the optimal classifier, hyperparameter tuning was again conducted throughout the best configuration with different numbers of estimators (50, 75), tree depths (3,4), learning rate (0.1, 0.2), and feature count (100, 150, 200, 300). At 100 features, the performance peaked and dropped considerably beyond 150 features, suggesting that higher feature counts caused overfitting.

2.8.7. Integrating Histological Grade as a Feature

Along with 121 selected phikon-v2 extracted features, histological grade was also included as the 1025th feature to see if it increases the model's performance. Tumor grade was incorporated as a binary variable (high = 1, low = 0). While feature selection using ANOVA F-statistics was applied to the 1024 phikon-v2 extracted features, grade was always kept as a constant input feature regardless of its F-score. It was done to ensure that the grade was treated as a clinical confounder instead of a competing feature.

2.8.8. LOO-CV as Cross-Validation Strategy

LOO-CV was applied as the assessment strategy since the sample size was very limited (77 patients only). It ensured that in each case, one patient was kept as a held-out test case and the remaining samples were used for training (Sun, 2025). All preprocessing steps, including feature scaling and selection, were performed independently within each fold using only the training data. This ensured that no data was leaked during the training process. The process was repeated 77 times until each patient had served as the test case exactly once. Along with that, StratifiedGroupKFold was applied to prevent data leakage ("StratifiedGroupKFold," n.d.).

2.8.9. Final Model Threshold

The final model was a Gradient Boosting Classifier from sklearn.ensemble package with 75 n_estimators, max depth = 3, learning rate = 0.1, selected features = 120 phikon-v2 + grade, scaler = StandardScaler, and Leave-One-Out as cross-validation ("GradientBoostingClassifier," n.d.) (Sun, 2025).

A classification threshold of 0.5 was applied to the predicted probabilities to obtain binary predictions. The model achieved a sensitivity of 0.9 and specificity of 0.43 at this threshold, correctly identifying 27 of 30 *TERT* promoter mutant cases while missing only 3.

2.8.10. Generation of a Grade-Only Model

To detect if the model is actually learning to detect the *TERT* promoter mutation instead of histological grade, or simply grade-associated tissue patterns, another model was implemented. This model used the identical model architecture, feature selection strategy, and LOO-CV protocol (Sun, 2025). This model was trained to predict histological grade from phikon-v2 image features alone. If the model were primarily encoding grade-related morphology, it would be expected to predict grade with similar or higher accuracy than TERT mutation status.

2.8.11. *TERT* Promoter Mutation Subtype Analysis

TERT promoter mutations predominantly occur at two hotspot positions. These are C228T and C250T. To evaluate whether the model demonstrated differential detection capability throughout the mutation subtypes, subtype-specific AUC analysis was conducted using the predicted scores from the final phikon + grade model using sklearn.metrics package of Python ("sklearn.metrics," n.d.).

To perform a subtype-specific comparison, estimated scores from the corresponding mutant subtype patients and all wild-type patients were obtained from the LOO-CV results using the numpy package of Python (Sun, 2025) (Harris et al., 2020). Then, the AUC was computed separately for C228T vs wild type and C250T vs wild type. The one patient with co-occurring mutations was excluded from this particular analysis.

2.8.12. Grade-Stratified Analysis

A grade-stratified analysis was conducted to see whether the model learns to detect p*TERT*-specific morphological alterations or if it is simply learning to detect histological grade instead. To do so, patients were divided into two groups based on their histological grade; these were high grade (n = 29) and low grade (n = 48). In this case, the phikon-v2 features were used with the LOO-CV pipeline, without grade as an input feature, and it was applied independently to each subgroup (Sun, 2025; Filiot et al., 2024). It allowed for determining whether p*TERT* mutation prediction performance was maintained when grade-related morphological variation was limited. In this case, the number of ANOVA F-statistic feature selection selected 50 features within each LOO-CV fold using only training data (Sun, 2025). The number of selected features was reduced to account for smaller stratified sample sizes.

2.8.13. Variant Allele Frequency Analysis

In the post-hoc evaluation framework, variant allele frequency (VAF) data acquired from tumor DNA ddPCR analysis were incorporated to examine the biological basis behind the grade-stratified performance differences and the false negative cases. VAF represents the percentage of tumor cells in a sample that contain the p*TERT* mutation ("GIL," n.d.). VAF values were available for all the patients in our dataset.

Here, using the VAF information, two analyses were conducted. At first, the distribution of VAF was compared across the high-grade and low-grade p*TERT* mutated subgroups using the Mann-Whitney U test from the Python package scipy.stats to identify whether the model's performance differences across the grade-based stratified groups could be interpreted by the differences in the fraction of mutated cells. The second analysis compared VAF and correctly detected mutant cases and false negative cases to evaluate if low allelic fraction contributed to failed detection.

Section 3

Results

3.1. Patient Demographics and Clinical Characteristics

The study comprised 77 patients with bladder cancer. The cohort predominantly consists of male patients, with 60 male patients, comprising 77.9% of the cohort, while the number of female patients was 17, comprising 22.1% of the dataset. The mean age of the male patients was 57 ± 11.9 years, and for the female patients, it was 53.8 ± 12.2 years, with median ages of 60 and 50 for males and females, respectively. The mean BMIs for the male and female patients were 22.12 ± 3.44 kg/m² and 20.89 ± 3.99 kg/m², respectively.

Hematuria was present in 66 patients, comprising 85.7% of the dataset. The percentage of recurrent patients was 50.6%, and primary patients were 49.4%, making for almost an equal distribution of both types. The dataset contains 29 high-grade tumors (37.7%) and 48 low-grade (62.3%) cases. Histopathological stage information was only available for 55 patients. Among them, 37 (48.1%) were categorized as pT1, and 18 (23.4%) were categorized as pT2. The remaining 28.6% had no stage information. Only 14 (18.2%) patients had type II diabetes mellitus, whereas the majority were non-diabetic.

3.2. Sensitivity of ddPCR for Detection of *TERT* Promoter Mutation

Through ddPCR analysis, *TERT* promoter detection was detected in 30 patients (39%). Among them, the C228T mutation was found in 22 patients (28.6%), and the C250T mutation was found in 7 patients (9.1%). Only one patient (1.13%) had both mutations. The rest of the 47 patients, comprising 61% of the dataset, were found to have a wild-type *TERT* promoter mutation. 9 out of 29 high-grade patients and 21 out of 48 low-grade patients harboured *TERT* promoter mutations.

The ddPCR method applied to tumor DNA samples successfully identified *TERT*p mutation across a varying range of allelic fraction. The VAF values spanned between 0.11% and 62.72% among the mutation positive samples. Also, three cases BD-UC-021, BD-UC-138, and BD-UC-151, showed very small varant allelic frequency of 0.71%, 0.85% and 0.11%, respectively. This phenomenon indicates that ddPCR is capable of detecting very low allelic fraction mutations and it is highly sensitive. Because of this superior property of ddPCR, data generated by using this method were used as the ground truth for our model development.

Chi-square or Fisher's exact test was applied according to the concordance to evaluate the association between clinical parameters and *TERT*p mutational status. The results are summarised in Table 3.3. 45% of male patients and 17.6% of female patients were *TERT*p

mutation positive. The association between gender and *TERT*p mutation showed borderline significance, with a p-value of 0.05, indicating a trend of higher *TERT*p mutational frequency in male patients than in female patients. The *TERT*p mutations were present in 43.8% of low-grade tumors, while comprising a frequency of 31% in high-grade cases, among grade subgroups. But this difference in *TERT*p mutation distribution across different grade subgroups was not statistically significant (p-value = 0.27) (Figure 3.3). Again, no significant association was found between *TERT*p mutational status and pathological stage, hematuria, recurrence status, and previous history of having kidney stones. Overall, *TERT*p mutation appeared as an early somatic alteration occurring irrespective of grade status in urothelial carcinoma.

3.3. Feature Extraction & Analysis using Phikon-v2

The 1024-extracted features from images using phikon-v2 were analysed using Spearman rank correlation with proxy features, revealing that most of the extracted features (n = 508, 49.61%) correlated strongly with edge-related morphological characteristics. The remaining 267 (26.07%) and 249 (24.32%) features were correlated to intensity and texture, respectively. This finding suggests that phikon-v2 largely encodes structural boundary information in H7e images (Filiot et al., 2024). This phenomenon is consistent with its pre-training on histological data.

3.4. Classifier Selection and Hyperparameter Optimisation

By using a systematic evaluation of multiple classifiers through feature counts of 20, 30, 45, 60, and 100, it was identified that the Gradient Boosting Classifier was the best-performing model. Using 100 Phikon-v2 features, Logistic Regression achieved a maximum AUC of 0.6191, Support Vector Machine achieved 0.6032, and Gradient Boosting achieved an AUC of 0.7213 (“1.4. Support Vector Machines,” n.d.; “GradientBoostingClassifier,” n.d.) (“LogisticRegression,” n.d.). Further hyperparameter optimisation revealed n_estimators = 75, max_depth = 3, learning_rate = 0.1 as optimal configurations, with 120 Phikon-v2 features plus grade information as the 121st feature. It achieved the highest AUC of 0.7674.

Table 3.1: Demographic information of bladder cancer patients.

Clinical Parameters and Characteristics of Bladder Cancer Patients (n=77)				
	Sex		Mean (±SD)	Median (IQR)
Age (years)	Male		57.0 (±11.9)	60 (19)
	Female		53.8 (±12.2)	55 (15)
BMI (kg/m²)	Male		22.12 (±3.44)	22.09 (5.54)
	Female		20.89 (±3.99)	20.36 (2.03)
Clinical Characteristics			Frequency (n)	Percentage (%)
Sex	Male		60	77.9
	Female		17	22.1
Hematuria	Yes		66	85.7
	No		10	13.0
	Missing		1	-
Recurrence	Recurrent		39	50.6
	Primary		38	49.4
Grade	High Grade		29	37.7
	Low Grade		48	62.3
Pathological Stage	PT1		37	48.1
	PT2		18	23.4
	Missing		22	28.6
Type II DM	Yes		14	18.2
	No		60	77.9
	Missing		3	-

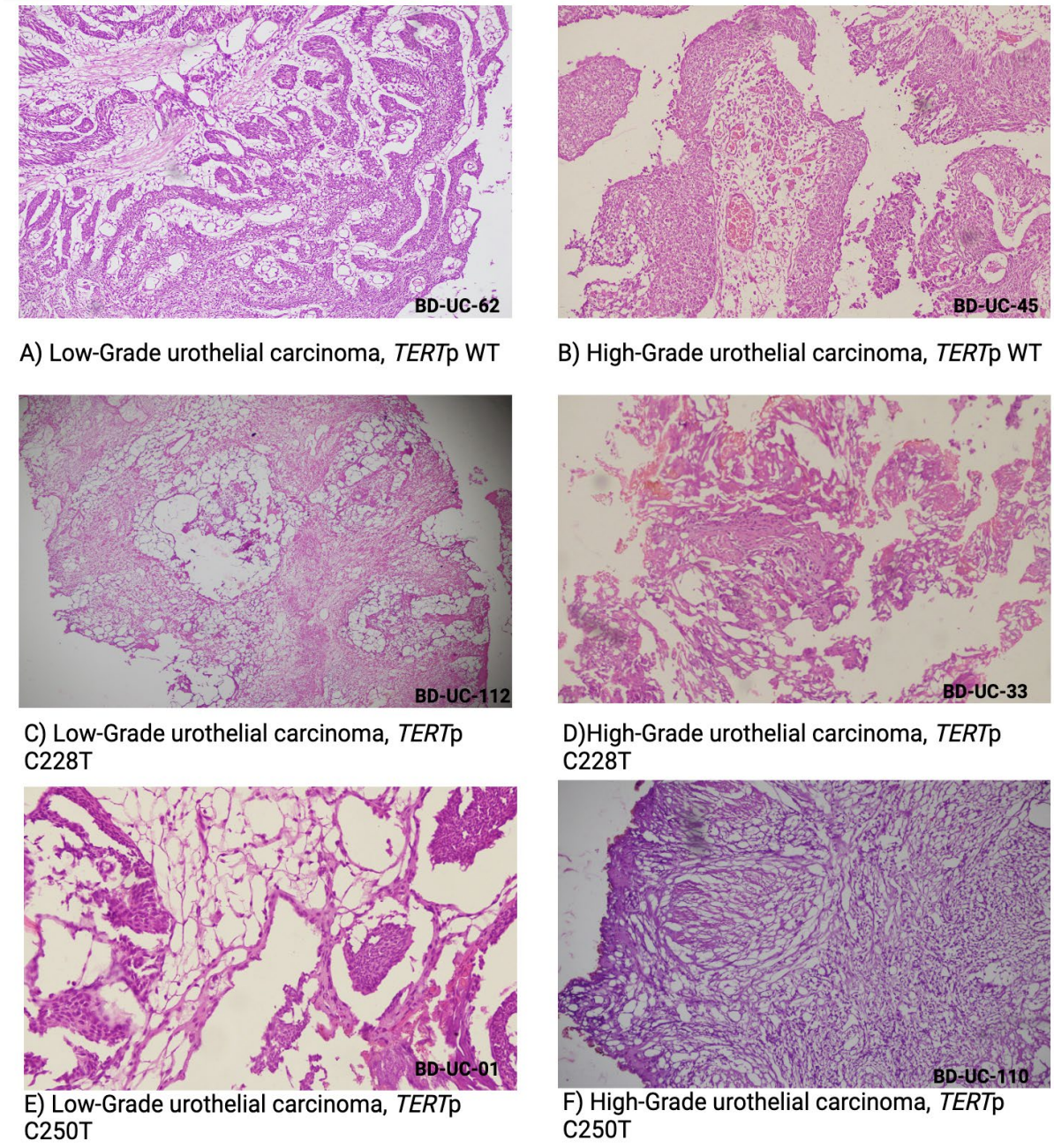


Figure 3.1: H&E-stained histological slides of bladder cancer. A) This was taken from a low-grade case where *TERT*p mutational status was wild-type. B) This one was from a High-grade case where *TERT*p mutational status was wild-type. C) This was from a low-grade case with C228T subtype of *TERT*p mutation. D)This was from a high-grade case with C228T subtype of *TERT*p mutation. E) This was from a low-grade case with C250T subtype of *TERT*p mutation. F) This was from a high-grade case with C250T subtype of *TERT*p mutation.

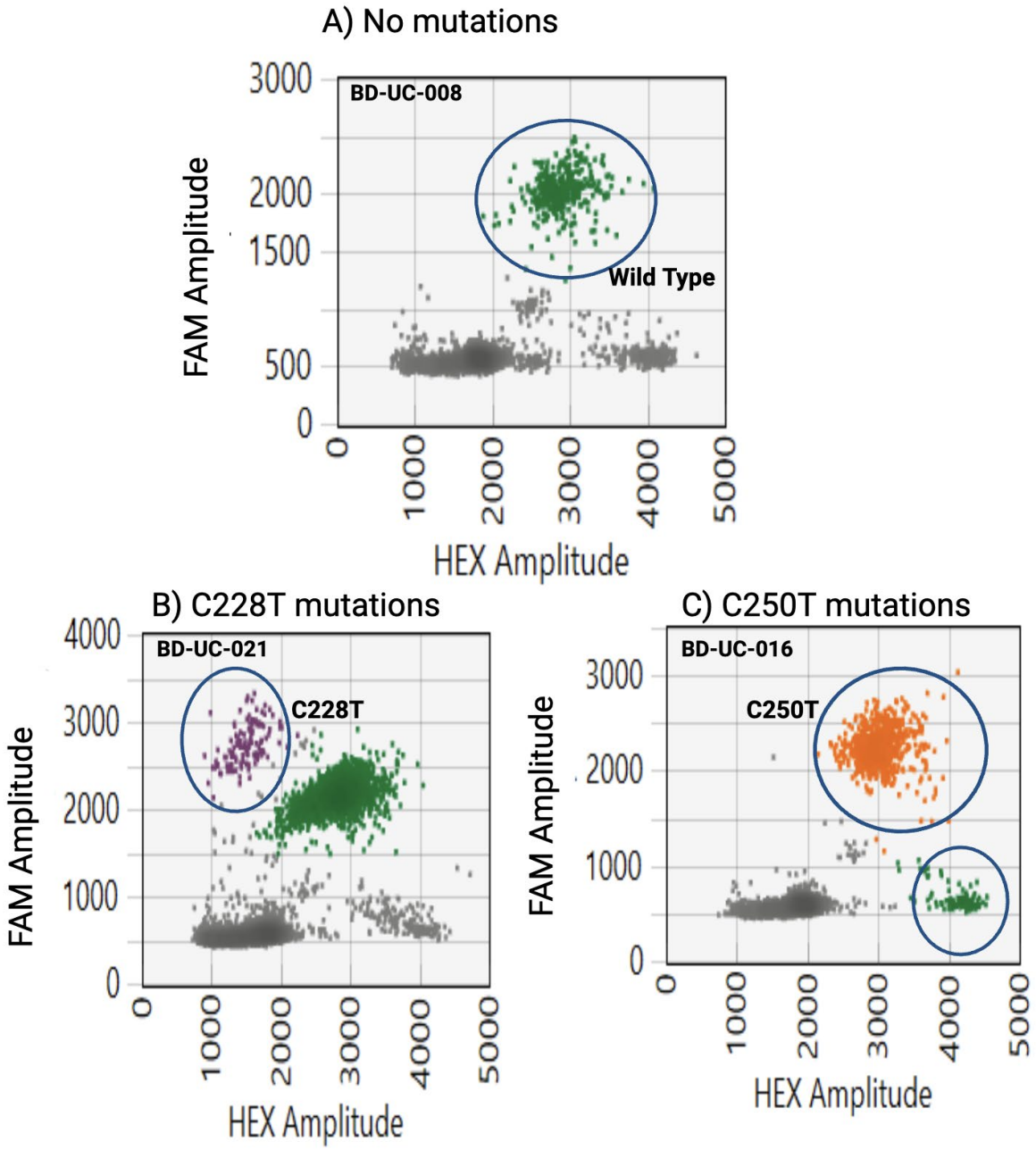


Figure 3.2: *TERT*p mutation from ddPCR analysis. A) A tumor sample with no *TERT*p mutation. B) A case containing C228T mutation and wild-type form of *TERT*p. C) A tumor containing C250T mutation and wild-type form of *TER*

Table 3.2: TERT Promoter Mutant Cases including Patient ID, Mutation Subtype and VAF

Patient ID	Mutation Subtype	VAF (%)
BD-UC-001	C250T (chr5:1,295,135 (C>T))	38.34
BD-UC-002	C250T (chr5:1,295,135 (C>T))	29.49
BD-UC-015	C228T (chr5:1,295,113 (C>T))	37.83
BD-UC-016	C250T (chr5:1,295,135 (C>T))	14.01
BD-UC-021	C228T (chr5:1,295,113 (C>T))	0.71
BD-UC-033	C228T (chr5:1,295,113 (C>T))	14.68
BD-UC-046	C228T (chr5:1,295,113 (C>T))	10.75
BD-UC-050	C228T (chr5:1,295,113 (C>T))	22.35
BD-UC-052	C228T (chr5:1,295,113 (C>T))	3.56
BD-UC-060	C228T (chr5:1,295,113 (C>T))	45.67
BD-UC-061	C228T (chr5:1,295,113 (C>T))	37.82
BD-UC-064	C228T (chr5:1,295,113 (C>T))	45.79
BD-UC-069	C250T (chr5:1,295,135 (C>T))	18.92
BD-UC-073	C228T (chr5:1,295,113 (C>T))	43.46
BD-UC-077	C228T (chr5:1,295,113 (C>T))	36.53
BD-UC-097	C228T (chr5:1,295,113 (C>T))	8.42
BD-UC-101	C250T (chr5:1,295,135 (C>T))	32.24
BD-UC-104	C228T (chr5:1,295,113 (C>T))	4.45
BD-UC-105	C228T (chr5:1,295,113 (C>T))	45.74

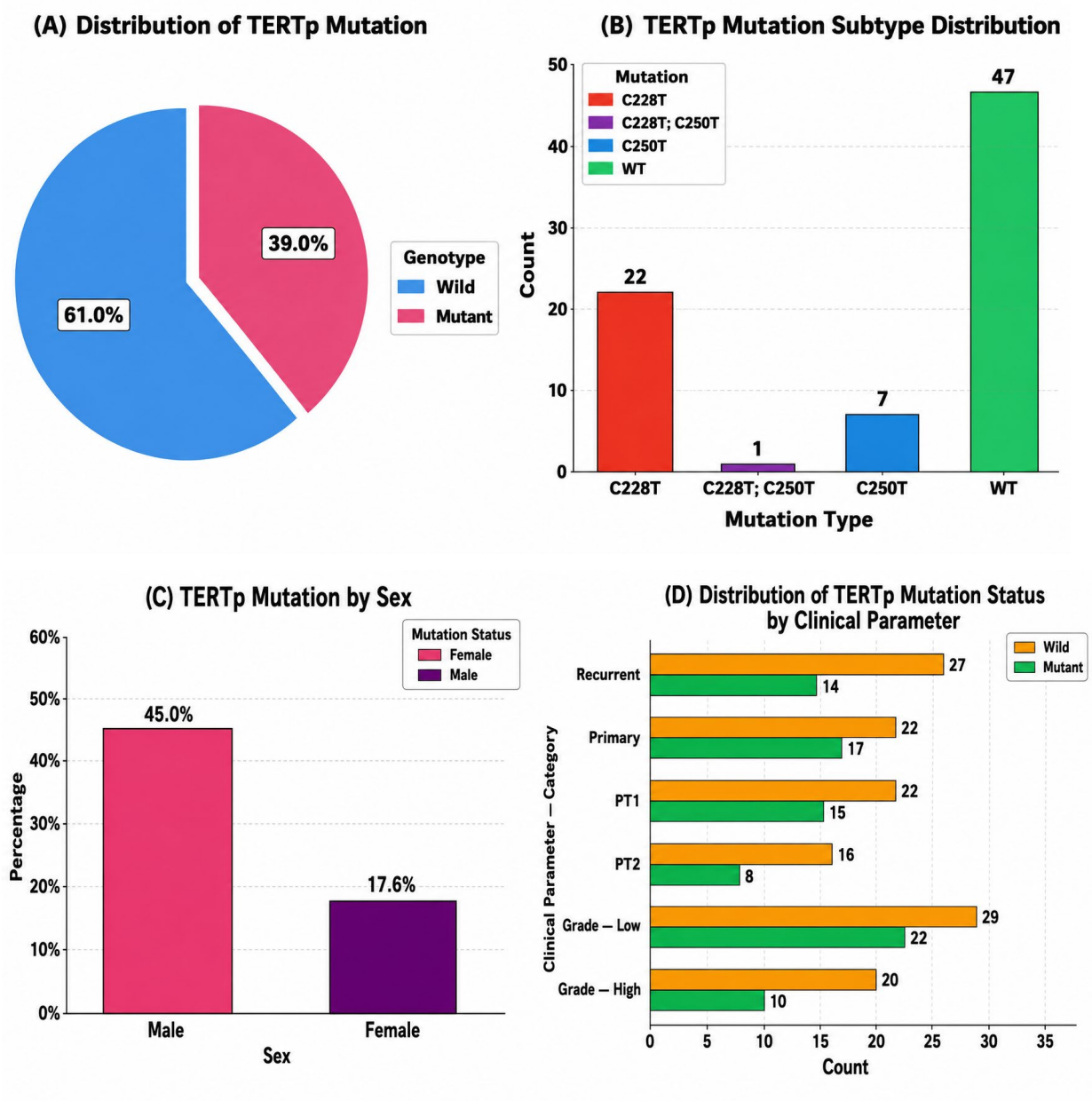


Figure 3.3: Distribution of *TERT* promoter mutation in our dataset. A) Pie chart showing the frequency of *TERT*p mutation across the cohort. B) Bar plot showing frequency of different subtypes of *TERT*p mutation and wild-type cases. C) Bar plot showing the frequency of *TERT*p mutation across sex. D) Bar plot showing distribution of *TERT*p mutation across different clinical parameters, such as recurrence history, histological stage, and grade

Table 3.3: Association of *TERT* promoter mutation with different clinical parameters.

Association between Clinical Parameters and TERTp Mutation Status (n=77)						
Clinical Parameter	**Total**	**Mutated**		**Wild Type**		**P value**
		n	**%**	**n**	**%**	
Sex	**Male**	27	45.0	33	55.0	0.05
	Female	3	17.6	14	82.4	
Grade	**High**	9	31.0	20	69.0	0.27
	Low	21	43.8	27	56.2	
Pathological Stage	**PT1**	15	40.5	22	59.5	0.36
	PT2	5	27.8	13	72.2	
Hematuria	**Yes**	27	40.9	39	59.1	0.30
	No	2	20.0	8	80.0	
Recurrence	**Recurrent**	15	38.5	24	61.5	0.93
	Primary	15	39.5	23	60.5	
Kidney Stone	**Yes**	3	33.3	6	66.7	>0.99
	No	22	36.1	39	63.9	

3.5. Model comparison

Three models were developed using Phikon-v2 features from H&E histological images as input (Filiot et al., 2024).

Table 3.4: The comparisons between these three models.

Model	**Input**	**AUC**
Image → Grade (control)	Phikon-v2 features	0.6279
Image → *TERT*p	Phikon-v2 features	0.7213
Image + Grade → *TERT*	Phikon-v2 + Grade	0.7674

The image-to-grade model was made to see if it learnt to detect grade over *TERT* promoter mutation. But the AUC was substantially lower than that of the image-to-*TERTp* model using only Phikon-v2-extracted features as input. The model's performance increased further when grade was added alongside the 120 ANOVA-selected Phikon-v2 features, yielding a final AUC of 0.7674 (Filiot et al., 2024) (Figure 3.4).

3.6. Final Model Performance

Gradient Boosting with 120 Phikon-v2 features + histological grade was selected as the final model, with an AUC of 0.7674 under LOO-CV evaluation (Filiot et al., 2024; Sun, 2025). At the standard classification threshold of 0.5, the model had correctly detected 27 out of 30 *TERTp* mutant cases (Figure 3.5).

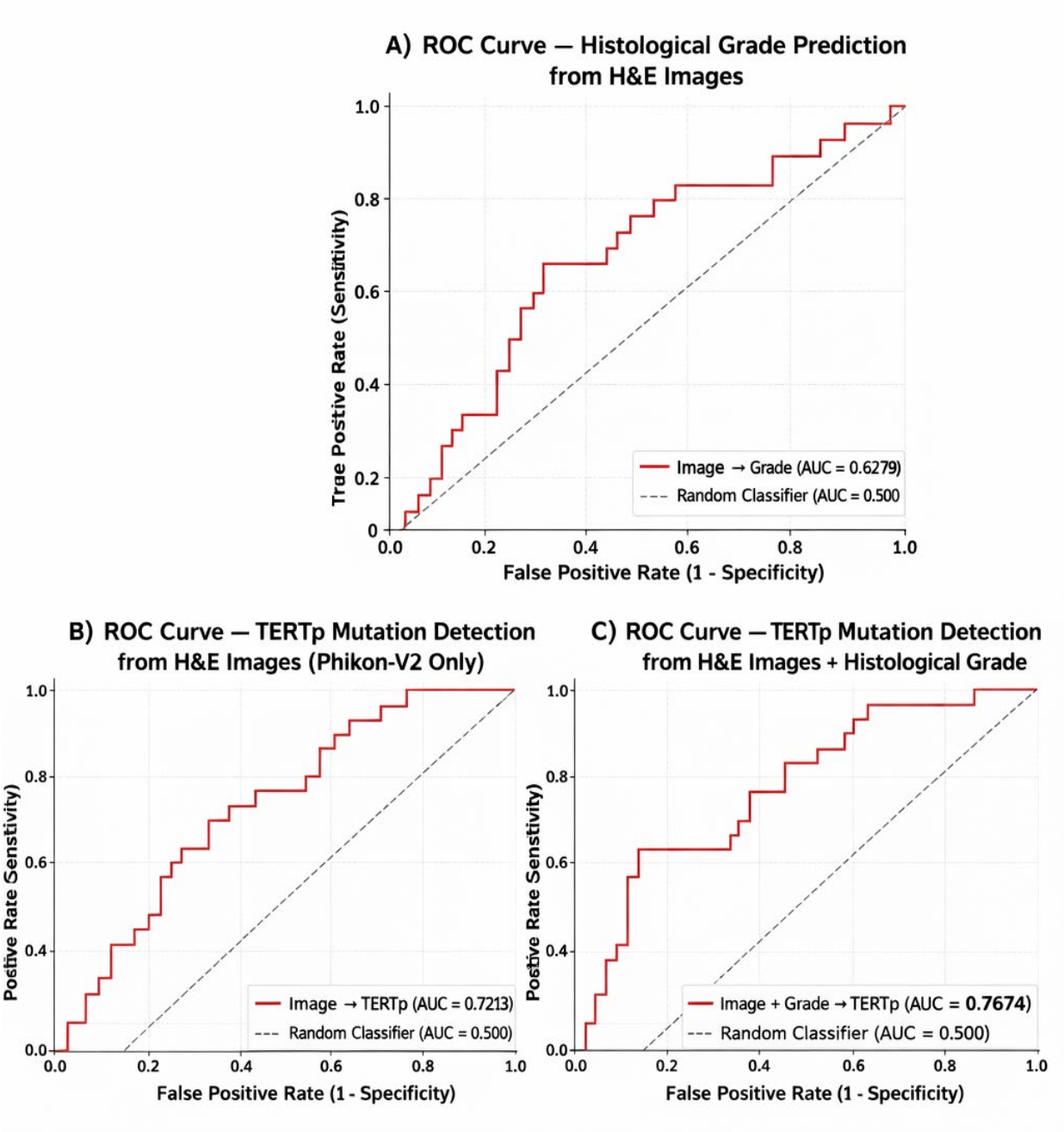


Figure 3.4: ROC curves for the comparison of the performances of the three models. A) The ROC curve for determining histological grade from image shows an AUC of 0.6279. B) The ROC curve for determining *TERT*p mutation from the image shows an AUC of 0.7213. C) The ROC curve for determining *TERT*p mutation from the image shows an AUC of 0.7674.

Table 3.5: The final model performance matrices to detect *TERT*p mutation from routine H&E-stained slide images.

Metric	**Value**
AUC	0.7674
Sensitivity	0.90
Specificity	0.43
Accuracy	0.61
True Positives	27
True Negatives	20
False Positives	27
False Negatives	3

At the classic standard threshold, the model correctly identified 20 patients as wild-type. This can result in a substantial 26% reduction in sequencing cost from unnecessary molecular testing.

3.7. *TERTp* Mutation Subtype Analysis

Subtype-specific analysis was performed to see which subtype the model performs better in (C228T or C250T). The model yielded an AUC of 0.7369 for C228T vs wild-type detection (n = 22 C228T, 47 WT) and an AUC of 0.8419 for C250T vs wild-type (n = 7 C250T, 47 WT) (Figure 3.6). The better AUC in C250T mutation detection may indicate that this subtype may generate more distinct morphological tissue alterations compared to C228T subtype, despite the smaller sample size.

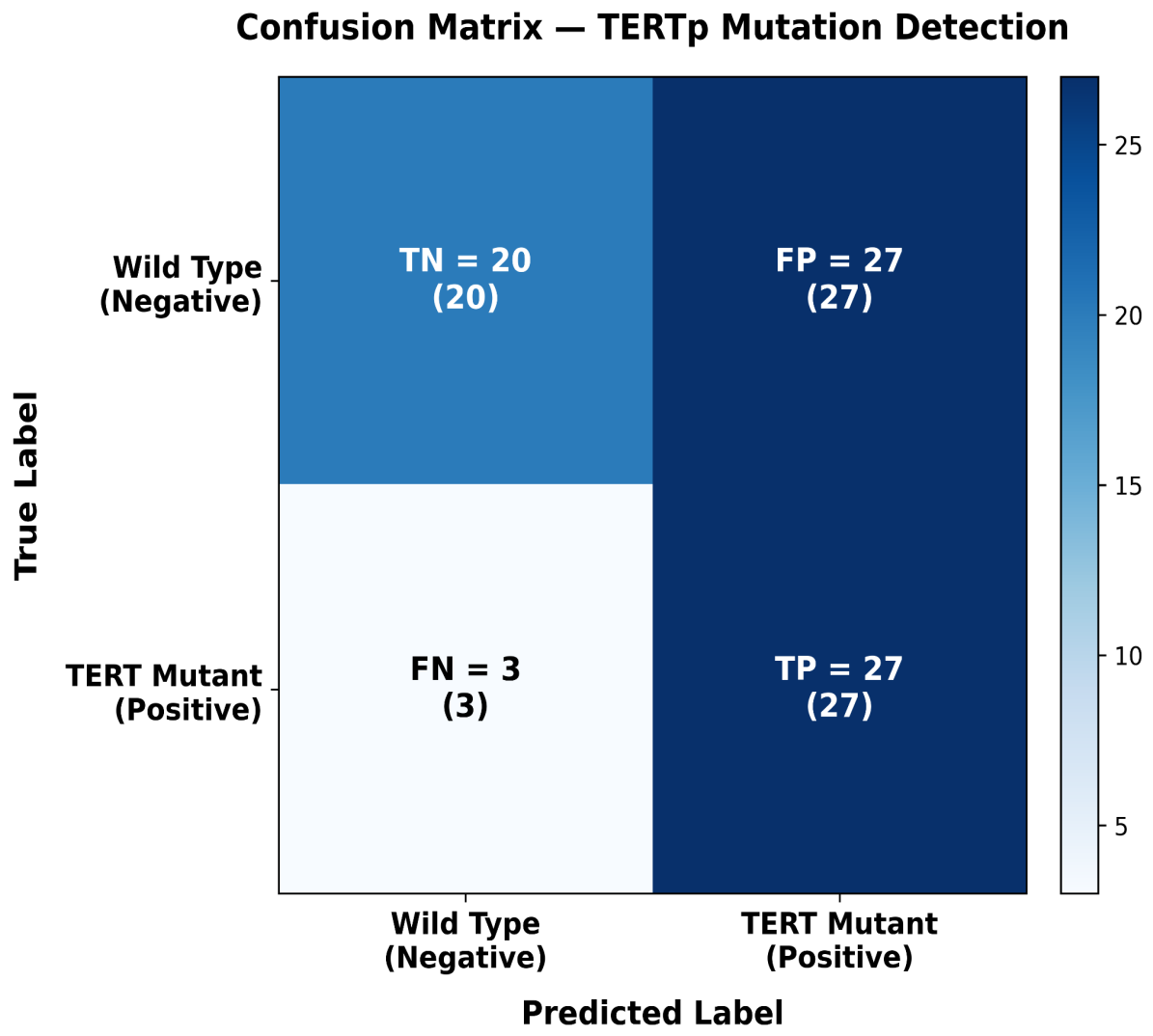


Figure 3.5: Confusion matrix for the final model. It depicts the overall performance of the model. It shows 20 true negative cases, 27 false positive cases, 3 false negative cases, and 27 true positive cases.

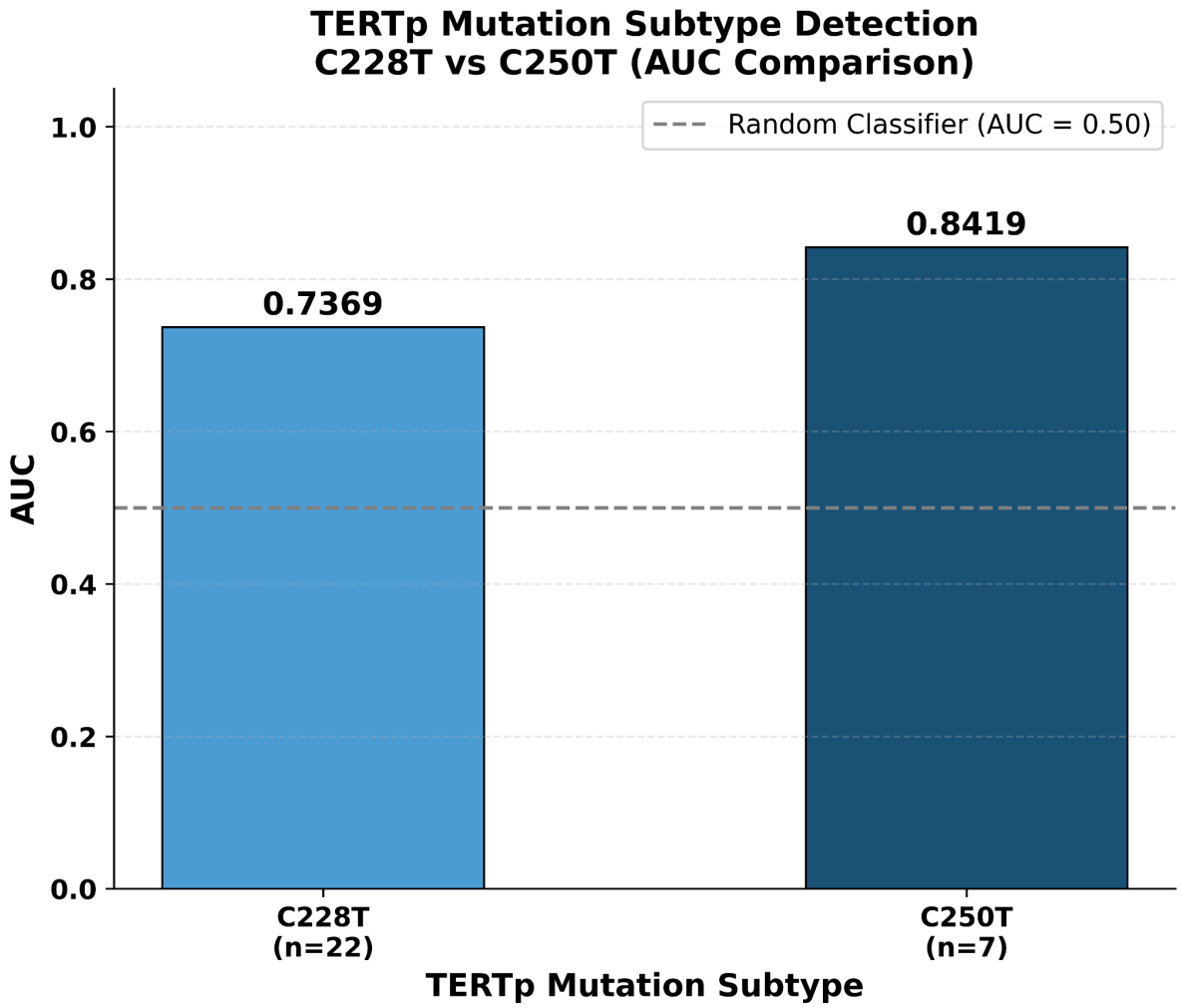


Figure 3.6: Bar plot showing performance differences of the model across the two subtypes. For the C228T subtype, the model gave an AUÇ of 0.7369. On the other hand, the model performed significantly better in the C250T subtype, giving an AUC of 0.8419.

3.8. Evaluation of Model through Grade-based Stratification

The datasets were split into High and Low grade and the model's performance across these two groups was evaluated. Using Phikon-v2 features alone, the high-grade-only model (n = 29) yielded an AUC of 0.7944 in the detection of the *TERT*p mutation (Filiot et al., 2024). In contrast, the low-grade-only group gave an AUC of 0.5256, suggesting near-random performance (Figure 3.7). The monomorphic features in low-grade tumors may contribute to this poor detection of the mutation from histological deep learning-based analysis.

These results indicate that, the overall achievement of an AUC of 0.7674 was mostly driven by the model's stronger detection capability in High-grade cases, where *TERT*p mutation may produce more distinct morphological changes in the already existing background of aggressive histological features. The failure to detect *TERT*p mutation in low-grade cases can be due to

low sample size and the monomorphic features of low-grade tumor itself making the histological alteration created by *TERT*p mutation too subtle to be detected by using deep learning.

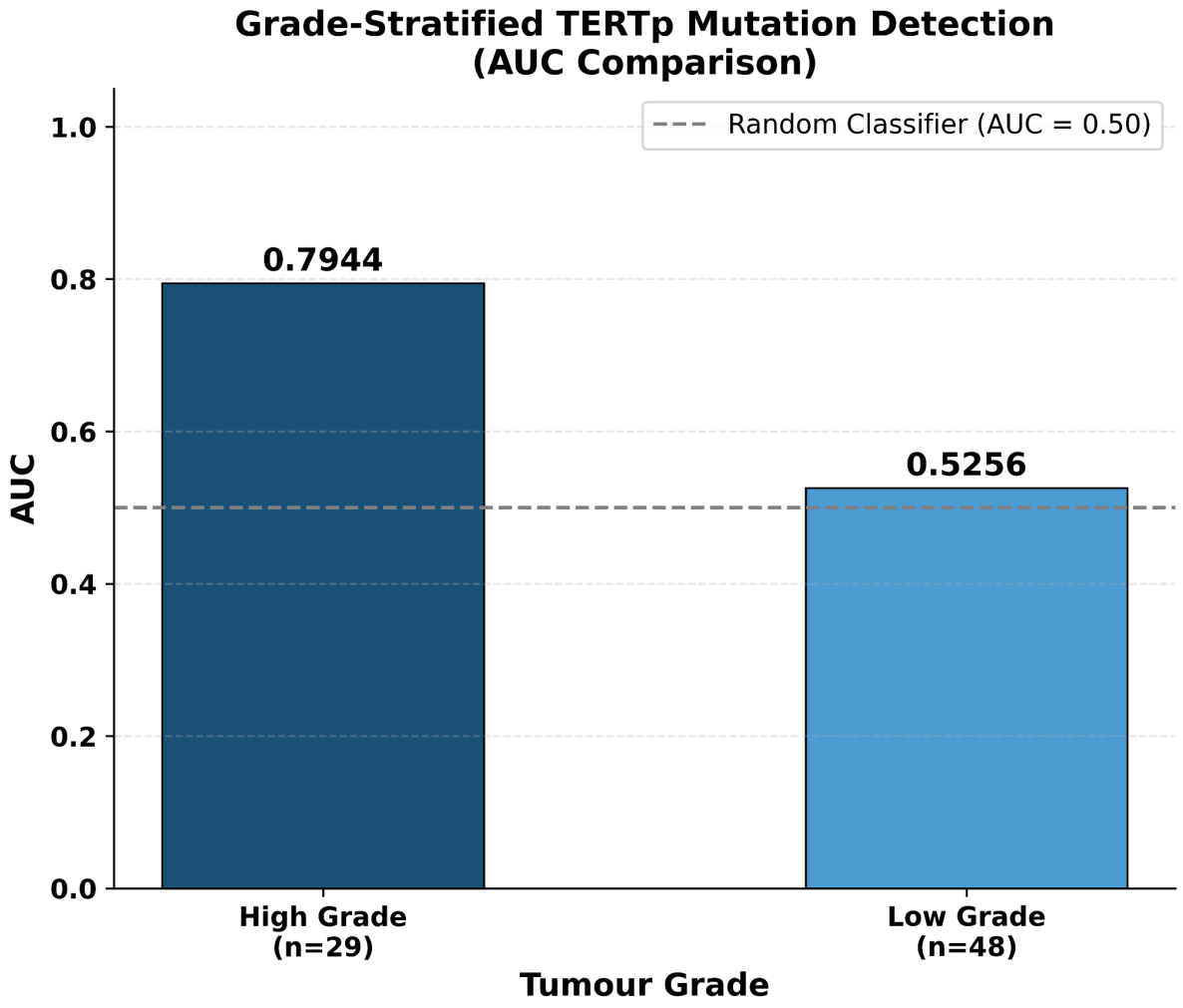


Figure 3.7: Bar plot showing performance difference of the model across different grades in grade-stratified analysis. The model gave an AUC of 0.7944 for the high-grade-only group, while it gave a near-random AUC in the low-grade subgroup.

3.9. Clinical Utility in terms of Reduction in Sequencing Cost

At the standard classification threshold, the model identified 23 out of 77 patients as wild-type, suggesting a potential 26% reduction in sequencing cost for this particular type of mutation. Although, the model missed 3 true mutants. To make this study usable as a pre-screening tool, validation on a different cohort to identify the sensitivity-prioritised threshold using Youden method, and subsequently using that threshold on another test cohort is necessary (Ruopp et al., 2008).

3.10. Association Between VAF & Model Detection Performance

In the first analysis, comparison of VAF between grade subgroups showed that low-grade *TERT*p mutant tumors had significantly higher VAF than the high-grade subgroup. The median was 36% for the low-grade, whereas it was only 14% for high grade. The Mann-Whitney U test gave a p-value of 0.024 (Chicco et al., 2025) (Figure 3.8 (A)). This finding suggests that the near-random detection performance for the low-grade subgroup in group-stratified analysis was not due to a reduced fraction of mutated cells; rather, the presence of a greater proportion of mutation across the low-grade cases probably made the *TERT*p-related morphological features very subtle for the model to detect in the monomorphic histological environment of low-grade urothelial carcinoma.

In the second analysis, the three false negative cases were found to have very low VAF compared to the correctly detected mutant cases (median 3% vs 29%) (Figure 3.8 (B)). At such a low allelic fraction, even though the mutation is present in a very small number of cells, it's likely producing too insignificant morphological changes to be detected by the model regardless of the tumor grade.

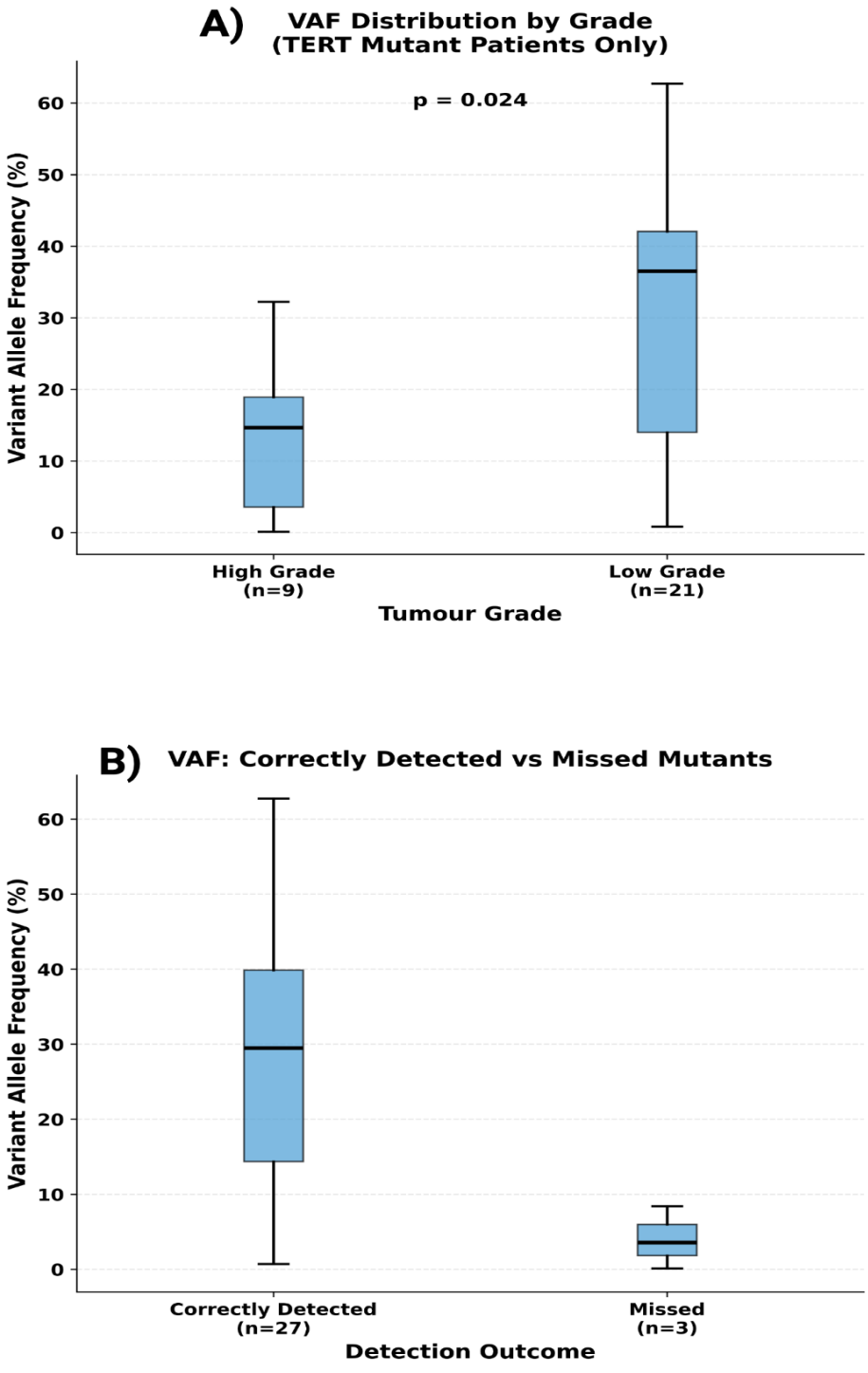


Figure 3.8: Box plots showing distribution of VAF across grade and correctly detected vs missed cases. A) This box plot demonstrated the distribution of VAF across high-grade and low-grade cases, where low-grade cases showed higher VAF values. B) This one depicts the distribution of VAF across correctly detected cases vs missed cases by the model, showing the correctly detected group had higher VAF values.

Section 4

Discussion

TERT promoter mutations at hotspot positions C228T(-124 bp) and C250T (-146 bp) upstream to the ATG start site are the earliest and most frequently occurring genetic alterations in urothelial carcinomas (Kurtis et al., 2016). Both of these mutations create a new consensus stretch of 11 bp nucleotides acting as a binding site for ETS family of transcription factors (Liu et al., 2013). This phenomenon results in 2-4 fold increase in *TERT* promoter activity and telomerase reactivation as a repercussion (Huang et al., 2013). Telomerase reactivation causes the stabilisation of short telomeres and enables cells to escape replicative senescence and results in early carcinogenesis of bladder cancer (Günes et al., 2018).

The observed frequency (39%) of *TERT*p mutation in this dataset is lower than the 50.85% reported in a recent study on bladder cancer in Bangladeshi population. Both of these study consistently detected C228T as the most prevalent mutation subtype. It accounted for 73.3% of mutation positive cases in this cohort and 70% in the previously reported study on Bangladeshi population (Ramadhan et al., 2025). *TERT*p mutation frequencies range from approximately 55.6% to 82.8% through different cohorts around the world using a plethora of detection methods (Jalkanen et al., 2023). The mutation frequencies observed in this and previously reported cohort fall within the lower end of the global range, indicating a potential population-specific difference in bladder cancer.

This study involved the collection of 77 tumor samples, all of which were from Bangladeshi population. The goal of this assessment was to detect *TERT*p mutation in both high and low allelic fractions using ddPCR and generate a pre-screening machine learning model to reduce unnecessary sequencing costs by detecting *TERT*p mutation from conventional H&E stained histology slides. For this reason, Phikon-v2, a pathology foundation model along with a Gradient Boosting classifier, was used (Filiot et al., 2024, v. 2; "GradientBoostingClassifier," n.d.). To our knowledge, till now, no such study has attempted to detect *TERT*p mutational status from routine H&E slides in bladder cancer. Some studies tried to detect*TERT*p mutational status from MRI images in glioblastoma but achieved a very low accuracy (49%) (Powter et al., 2021). The study tried to address the clinically relevant gap in resource-constrained settings where molecular testing facilities are not feasible. Two complementary approaches were generated. At first, the sensitivity of ddPCR in detecting low allelic fraction mutations was established, and secondly, the data generated from that analysis were used as a ground truth for our foundational pre-screening model to detect the *TERT*p mutation from conventional H&E-stained histology slides.

For this study, ddPCR was used to detect *TERT*p mutation from tumor DNA samples. This strengthened the methodological rigor of the study since conventional sequencing methods like Sanger sequencing are not appropriate enough to detect low allelic fraction mutations, and have a reported sensitivity of approximately 15-20% VAF. Below this threshold, mutations in heterogeneous tumor samples are usually undetectable through the Sanger sequencing method (Davidson et al., 2012). On the contrary, in this cohort, ddPCR detected three *TERT*p mutant cases that had VAF lower than 1%. This suggests that the mutations that would have been flagged as wild-type in the Sanger sequencing method had been correctly detected by ddPCR. This phenomenon indicates that ddPCR was a better choice for the ground truth generation in this analysis. The wide VAF range observed (0.11% to 62.72%) demonstrated the biological variability of *TERT*p mutation across urothelial carcinoma, emphasizing the necessity of high-sensitivity molecular methods and in-depth mutation characterisation.

The final model achieved an AUC of 0.7674 under LOO-CV evaluation, where 120 Phikon-v2 extracted features and grade information were used as the input (Filiot et al., 2024). The model achieved a sensitivity of 0.90 and a specificity of 0.43 at the standard classification threshold of 0.5. This finding is consistent with the fact that *TERT* promoter mutations are noncoding mutations that alter the regulatory control of telomerase transcription by activating it without directly changing the cellular architecture (Liu et al., 2013). Therefore, it is not expected for this mutation to generate morphological changes that could be visible on H&E images. But a few studies suggested that these changes can be detected by using deep learning methods. The AUC of 0.7674 is also comparable to other published studies predicting somatic mutations from H&E slide images in other cancer types, that ranges from 0.65-0.80 (Kather et al., 2020). Phikon-v2 extracted 1024 features from each image as a CLS token. The proxy feature analysis revealed that approximately 50% of the extracted features correlated strongly with edge-related morphological properties. This is consistent with the model's pretraining on histopathological patches, where nuclear and structural boundaries were the main discriminative features. It suggests that the model translates relevant morphological information for tissue characterization and a biologically reasonable framework for mutation prediction from H&E-stained images (Filiot et al., 2024).

A major concern of this study was whether the model was learning to detect grade instead of *TERT*p mutation, since our dataset contained both high and low-grade cases and their proportions were not similar. To assess this concern, a detailed confounder evaluation approach was applied. This comprised three complementary analyses documented in the computational

pathology literature. These were adjustment of confounders by incorporating grade as a definite model input, additional model comparison through independent grade and *TERT*p detection, and finally, grade-stratified evaluation within uniform grade subgroups ("Confounding factors and biases abound when predicting molecular biomarkers from histological images | Nature Biomedical Engineering," n.d.).

The image-to-grade prediction model achieved an AUC of 0.6279, whereas the Phikon-v2 features-only model achieved 0.7333 in mutation detection. It showed that Phikon-v2 features encode more robust discriminative information for *TERT*p mutation detection than for histological grade (Filiot et al., 2024). It suggests that the model captured morphological characteristics specific to *TERT*p mutation rather than grade-correlated tissue architectures. This observation was consistent with the established biology of *TERT*p mutation in bladder cancer, where the mutation occurs irrespective of histological grades, with reported frequencies of 73.7% in low-grade and 63.3% in high-grade bladder cancer (Cheng et al., 2023; Huang et al., 2013).

The addition of histological grade as an additional input improved the model performance from 0.7333 to 0.7674. As histological grade is routinely assessed as standard clinical practice during H&E slide examination, its inclusion does not require any additional workload or cost, and made the combined model more applicable as a pre-screening tool to reduce the sequencing burden.

From the grade-stratified analysis, it was revealed that the model substantially performed differently across different grade subgroups. While it showed a strong detection capability in the high-grade subgroup (AUC = 0.7944), it achieved a near-random AUC (0.5256) in the low-grade subgroup. This finding requires careful interpretation.

In low-grade bladder cancer, a relatively uniform architecture along with minimal cytological atypia and low mitotic activity are observed. Thus, it can be assumed that it provides limited morphological contrast between *TERT*p mutant vs wild-type tumors (Zaleski et al., 2019). Also, low-grade papillary urothelial carcinoma usually exhibits minimal loss of polarity at higher magnification and appears cytologically more monomorphic than the high-grade cases that demonstrate more pleomorphic structures ("Low-Grade Papillary Urothelial Carcinoma | Basicmedical Key," n.d.). The differential model performance across different grade-stratified subgroups was plausibly caused by this morphological contrast across different histological grades. Moreover, since *TERT* promoter mutations are noncoding mutations occurring in the regulatory regions and transcriptionally activate the telomerase gene, it can be assumed that

even though the morphological alterations caused by them existed, it probably be masked by the already existing uniform morphological background in low-grade cases and led to the model's poor performance in this stratum.

VAF analysis revealed that the low-grade mutant cases had an overall higher VAF compared to the high-grade mutant cases (median 36% vs 14%, $p = 0.024$). It implies that the near-random performance in the low-grade subgroup was not due to a lower mutant cell fraction; rather, the presence of a high mutant cell burden. It plausibly explains the reason that the detection failure was more likely related to the morphological obscurity produced by the *TERT*p mutation in the already existing bland histological environment in low-grade urothelial carcinoma instead of a diluted mutant signal.

This overall outcome aligned with the established difficulty of morphology-based detection in low-grade bladder cancer, where even urine cytology achieves a sensitivity as low as 4% due to the monomorphic architecture of low-grade tumor cells (Li et al., 2013).

In the high-grade stratum, the tumours likely reflected enhanced *TERT*-related histological alterations. This is also amplified by the co-occurring mutations and the generally higher tumour mutational burden in high-grade cases, where multiple genetic alterations cause more extensive and detectable morphological alterations. Studies on pan-cancer mutation prediction using deep learning have revealed that mutations are more robustly detected from histology in tumours with higher mutational burden ("Self-supervised attention-based deep learning for pan-cancer mutation prediction from histopathology | npj Precision Oncology," n.d.).

However, the high-grade subgroup is very small, and only 9 cases were *TERT*p-mutant. So these findings are only hypothesis-generating and should be interpreted cautiously in this study.

A subtype-specific analysis was conducted, and it revealed differential detection performance across *TERT*p mutation subtypes. The model achieved a higher AUC of 0.8419 for C250T vs wild-type compared to an AUC of 0.7369 for C228T vs wild-type. The more robust detection of C250T mutations in spite of the smaller sample size ($n = 7$) may indicate that this mutation could likely create more distinct morphological changes than C228T. While both C228T and C250T mutations create de novo ETS transcription factor binding sites activating *TERT* expression, they follow different mechanisms. The C228T activates *TERT* by binding the ETS alone, and on the contrary, C250T does so by a non-canonical NF-κB signalling in addition to ETS binding. The NF-κB signalling causes inflammation, immune infiltration, and tissue remodeling; all these are visible on H&E images (Lloyd and Sahai, 2015) (Y. Li et al., 2015b).

The involvement of NF-κB signalling inC250T subtypes may have contributed to its better detection compared to the C228T subtype.

The model achieved a sensitivity of 0.90 by correctly detecting 27 out of 30 *TERT*p mutant cases, at the standard classification threshold. The model would have filtered out 20 wild-type cases from redundant sequencing, representing a 26% reduction in molecular testing workload, under the study conditions. In resource-constrained environments, where sequencing and ddPCR facilities are not readily available, such a pre-screening approach could considerably reduce the number of samples requiring molecular testing while maintaining clinically acceptable mutant detection rates.

The three missed mutant cases had significantly low VAF values compared to the correctly identified ones (median 3% vs 29%). It indicates that, at a very low allelic fraction, where the mutation is present in only a small fraction of cells, the histological contribution of the *TERT*p mutation may fall below the detection threshold of the model irrespective of the tumor grade.

The classification threshold used in this study was applied at the standard value of 0.5. The threshold optimization to achieve a near-perfect sensitivity for clinical implementation should be conducted on an independent validation cohort ("AI prediction models based on time-lapse imaging for good embryos with implantation potential and euploidy | Scientific Reports," n.d.). The current performance metrics represent cross-validated results from a single institution cohort, and it mandates external validation before clinical deployment.

In addition to cost reduction, this proposed pre-screening model can serve other objectives of personalized medicine and precision oncology in the management of bladder cancer. Also, *TERT*p mutational status has been known to emerge as a clinical biomarker, and it is associated with disease recurrence and worse prognosis (Rachakonda et al., 2013). Thus, tailored treatment planning and enhanced surveillance could be possible from initial diagnosis of *TERT*p mutation from histological assessment without demanding expensive additional molecular testing settings. A conventional H&E-stained image-based pre-screening tool like this could be used as a first line triage step in resource-limited settings like Bangladesh, where ddPCR and NGS remained restricted due to their high cost and infrastructural limitations. Integration of this type of tool along with the already existing conventional diagnosis method may reduce the financial burden on both healthcare systems and patients, while making personalised medicine more widely accessible.

In this study, no held-out test was performed due to the limited sample size consisting of only 77 patients. In the case of a dataset with a low sample size, it is generally discouraged to

partition the data into separate training, validation, and test cohorts because it substantially reduces the number of samples available for training the model ("On Splitting Training and Validation Set: A Comparative Study of Cross-Validation, Bootstrap and Systematic Sampling for Estimating the Generalization Performance of Supervised Learning | Journal of Analysis and Testing | Springer Nature Link," n.d.; vutukuri, 2025). That is why Leave-One-Out-Cross-Validation (LOO-CV) was employed, as it maximizes the number of available samples for the training cohort (Sun, 2025). This type of internal validation is also recommended when the sample size is small ("Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results | PLOS One," n.d.). Therefore, the reported AUC of 0.7674 acts as an internal cross validated estimate.

There are several limitations of this study that need to be acknowledged. First, the limited sample size of 77 patients from a single institution in Bangladesh constrains the statistical power and the generalisability of the findings. The small subgroup sizes in grade-stratified and subtype-specific analyses may particularly introduce considerable variability in performance metrics. Secondly, the images were captured using an Olympus DP22 digital camera at 20x magnification rather than a digital whole slide imaging scanner. This resulted in a single low-resolution image per patient instead of acquiring multiple patch WSIs for which Phikon-v2 was originally designed (Filiot et al., 2024). Thirdly, while a detailed confounder assessment framework was employed, complete disentanglement of *TERT*p mutation-specific and grade-associated morphological signals was not achievable, and thus the contribution of grade confounding can not be fully excluded. Finally, no external cohort was available, and all metrics were obtained from a single institution dataset.

Future work should focus on increasing the cohort size. If acquiring WSIs is not possible, multiple images at different magnifications per patient should be obtained. Threshold optimization should be performed on a different cohort, and the model's performance should be evaluated on a test cohort for a generalisability check. The employment of a dedicated confounder-adjusted learning method, such as adversarial training or invariant risk minimisation, should be applied to further improve the *TERT*p-specific feature learning.

Section 6
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Section 6
Appendices

6.1.Consent form for study participants

- ○ Primary
- ○ Recurrent

- ○ CBC
- ○ Histopathology
- ○ USG
- ○ Biochemical

রোগীর নমুনা এবং তথ্য সংগ্রহের জন্য সম্মতিপত্র

আমি ..... (রোগীর নাম), আমার সজ্ঞানে নিম্নলিখিত গবেষণা প্রকল্পের জন্য আমার নমুনা (রক্ত, থুতু, টিউমার টিস্যু, মূত্র) ও তথ্য দিতে সম্মতি প্রকাশ করছি।

প্রকল্পের নাম: "Bioinformatics-Driven Discovery and Laboratory-Based Validation of a Six-Gene Signature as a Prognostic Biomarker in Bladder Cancer"

প্রকল্প পরিচালকবৃন্দ:

১. ডঃ মোঃ ইসমাইল হোসেন,
সহযোগী অধ্যাপক, প্রাণরসায়ন ও অনুপ্রাণ বিজ্ঞান বিভাগ, ঢাকা বিশ্ববিদ্যালয়

২. ডঃ হোসেন উদ্দিন শেখর
অধ্যাপক, প্রাণরসায়ন ও অনুপ্রাণ বিজ্ঞান বিভাগ, ঢাকা বিশ্ববিদ্যালয়

৩. ডঃ ফ্লোরেন্স লু কালভেজ-কেলম
বিজ্ঞানী, ইন্টারন্যাশনাল এজেন্সি ফর রিসার্চ অন ক্যান্সার, বিশ্ব স্বাস্থ্য সংস্থা, ফ্রান্স

প্রকল্পের ক্লিনিক্যাল সহযোগী পরিচালক:

১. ডাঃ এইচ ফজলুল করিম, সহযোগী অধ্যাপক ও বিভাগীয় প্রধান, ইউরো অনকোলজী, জাতীয় ক্যান্সার গবেষণা ইনস্টিটিউট ও হাসপাতাল, ঢাকা

আমি এই গবেষণা শেষে প্রকাশনার জন্য আমার তথ্য (নাম ছাড়া) ব্যবহারের অনুমতি দিচ্ছি।

১। রোগীর নাম: .....

২। তারিখ: .....

রোগীর স্বাক্ষর: .....

6.2.

প্রকল্পের নাম: "Bioinformatics-Driven Discovery and Laboratory-Based Validation of a Six-Gene Signature as a Prognostic Biomarker in Bladder Cancer"

Patient Admission Number: .....

সনাক্তকরণ নম্বর: UBC- |__|__|__|__|

আমাদের প্রকল্পে অংশগ্রহণে সম্মত হওয়ার জন্য আপনাকে অনেক ধন্যবাদ।
এই গবেষণা চালিয়ে যাওয়ার জন্য, আমাদের আপনার জীবনধারা সম্পর্কে কিছু তথ্য সংগ্রহ করতে হবে। আপনি যদি একমত হন, আমি আপনাকে কিছু প্রশ্ন জিজ্ঞাসা করব এবং এই ফর্মটিতে এই সম্পর্কে আপনার উত্তর লিখব। এতে প্রায় ১০ মিনিট সময় লাগবে। যদি সম্ভব হয় সঠিক উত্তর দিয়ে আমাদের সাহায্য করবেন।

১। রোগীর নাম: .....

২। প্রশ্নপত্র পূরণের তারিখ: |__|__| / |__|__| / |__|__|__|__
(দিন / মাস / বছর)

৩। তথ্য সংগ্রহকারীর নাম এবং পদবী :

সাধারণ তথ্য

৪। লিঙ্গ : (1) পুরুষ (2) মহিলা

৫। বয়স : ..... বছর

৬। শিক্ষা :

৭।ক। আপনার উচ্চতা ? .....

খ। বর্তমানে আপনার ওজন কত ? |__|__|__| কেজি

৮।ক। স্থায়ী ঠিকানা:

খ। বর্তমান ঠিকানা:

৯। রক্তচাপ: .....

১০। ফোন নম্বর : .....

6.3.Questionnaire for collecting information on bladder cancer study participants

মেডিকেল ব্যাকগ্রাউন্ড

৯। আপনার কি কখনও নিম্নের অসুখ হয়েছে (এক বা একাধিক) ? :

ক। রোগবিদ্যা	খ। প্রথম লক্ষণ দেখা দেবার বয়স	গ। ডাক্তার নির্ধারিত চিকিত্সা নিচ্ছেন ?	ঘ। কি চিকিত্সা এবং কত বছর চিকিত্সা গ্রহণ করেছেন ?
অন্যান্য ক্যান্সার ? ☐ হাঁ ; কি ক্যান্সার ? .... ☐ না ☐ জানি না	___	☐ হাঁ ☐ না ☐ জানি না	**অস্ত্রোপচার হলে এইখানে টিক দিন ☐** **কতবার হয়েছে:** **মধ্যবর্তী সময়কাল:**
ডায়াবেটিস টাইপ 2 ☐ হাঁ ☐ না ☐ জানি না	___	☐ হাঁ ☐ না ☐ জানি না	
বারবার ইউরিন ইনফেকশন হয়েছে? (মূত্রাশয় পুনরাবৃত্তি ব্যাকটেরিয়া সংক্রমণ) **বা হেম্যাটো ইউরিয়া ?** ☐ হাঁ ☐ না ☐ জানি না	___	☐ হাঁ ☐ না ☐ জানি না	
কিডনি পাথর ☐ হাঁ ☐ না ☐ জানি না	___	☐ হাঁ ☐ না ☐ জানি না	

আপনার স্বাস্থ্য সম্পর্কে আরও যেসব তথ্য আপনি উল্লেখ করতে চান :

পরিবার ইতিহাস

১০। আপনার প্রিয়জনের (ভাই, বোন, পুত্র, কন্যা বা বাবা) কখনও ক্যান্সার হয়েছে? (সব ধরনের ক্যান্সার অন্তর্ভুক্ত)?

☐ হাঁ ☐ না

যদি হ্যাঁ হয় দয়া করে কার হয়েছে এবং কি ক্যান্সার হয়েছে সেটি লিখুন।

.....

যদি কেউ ক্যান্সারে মারা গিয়ে থাকেন সেটাও লিখুন দয়া করে।

.....

তামাক বা পান জর্দা গ্রহণ

১০.১ আপনি কি কখনো আপনার জীবনকালে ধূমপান ( ) বা পান-জর্দা ( ) সেবন করেছেন? ☐ হাঁ ☐ না

১০.২ কোন বয়সে আপনি শুরু করেন ? |__|__| বছর

১০.৩ আপনি কি এখনো করেন? ☐ হাঁ ☐ না (..... বছর আগে ছেড়েছেন)।

অ্যালকোহল এবং অন্যান্য পানীয় খরচ

১১। আপনি কি কখনও নিয়মিত অ্যালকোহল বা মদজাতীয় পানীয় পান করেছেন (অন্তত এক সপ্তাহের জন্য এক বছরের জন্য)? ☐ হাঁ ☐ না

পেশাগত রিস্ক

১২। আপনি কি কখনও নিম্নলিখিত ব্যবসা বা শিল্পে কাজ করেছেন? ? আপনার উত্তর যদি হ্যাঁ হয়, তবে জড়িত সময় এবং আপনি যেকোন রাসায়নিকের সংস্পর্শে আসলে লিখুন :

ব্যবসায় / শিল্প :	কত বছর কাজ করেছেন?	রাসায়নিক যৌগ (যদি জানা থাকে)
প্লাস্টিক বা রাবার শিল্প ☐ হাঁ ☐ না ☐ জানি না		
টেক্সটাইল শিল্প বা ডাইং শিল্প ☐ হাঁ ☐ না ☐ জানি না		
ছাপাখানা বা পেইন্টিং ☐ হাঁ ☐ না ☐ জানি না		
মেটাল শিল্প ☐ হাঁ ☐ না ☐ জানি না		
তেল কারখানা ☐ হাঁ ☐ না ☐ জানি না		

রোগীর পেশাঃ

আপনার মূল্যবান সময় দেবার জন্য আপনাকে অনেক ধন্যবাদ।

রোগীর স্বাক্ষরঃ .....

6.4. Certificate of Ethical Approval

National Institute of Cancer Research and Hospital
Mohakhali, Dhaka-1212.

Ref. no: NICRH/Ethics/2021/ 181 Date: 21·06·2021

Certificate of Ethical Approval

Title: "Investigating the potential of urinary TERT promoter mutations as a non-invasive biomarker for bladder cancer in Bangladesh using ddPCR approach (Bangla-TERT)."

Investigator:

Recommendation: APPROVED/~~APPROVED AFTER CORRECTION/NOT APPROVED~~

Note: The research study to be carried out must comply with the national laws and regulations of the country and "WMA declaration of Helsinki-Ethical Principles for Medical Research Involving Human Subjects, amended, October 2008'

Prof. (Dr.) Qazi Mushtaq Hussain
Chairperson
Ethics Committee
NICRH

Dr. Mohammad Sayeed Hossain
Associate Professor
Member Secretary
Ethics Committee, NICRH

6.5. Plagiarism Report

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